

Interferometric imaging

ICCP 2026 Summer School

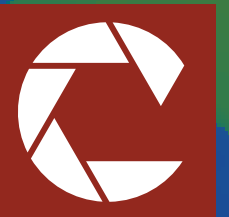
Sreekar Ranganathan

Mian Wei

Ioannis Gkioulekas

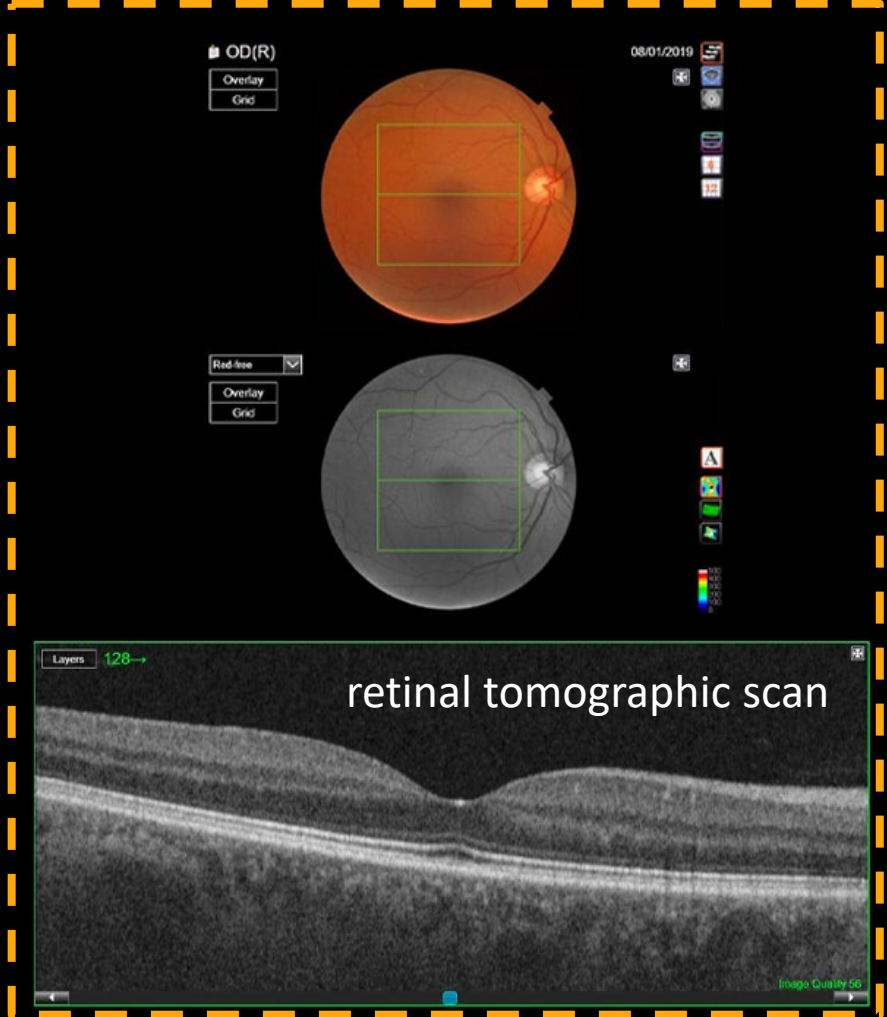
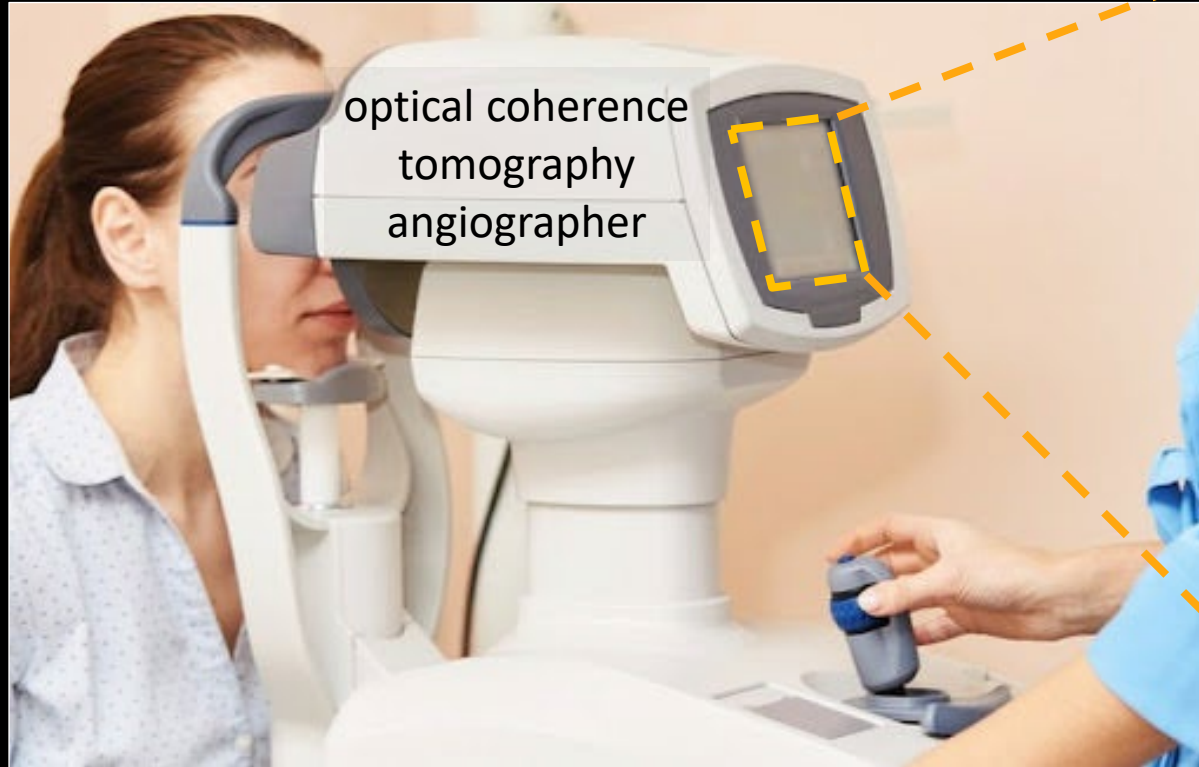
Carnegie Mellon University

<https://imaging.cs.cmu.edu/>



What is interferometry?

OCT eye exam

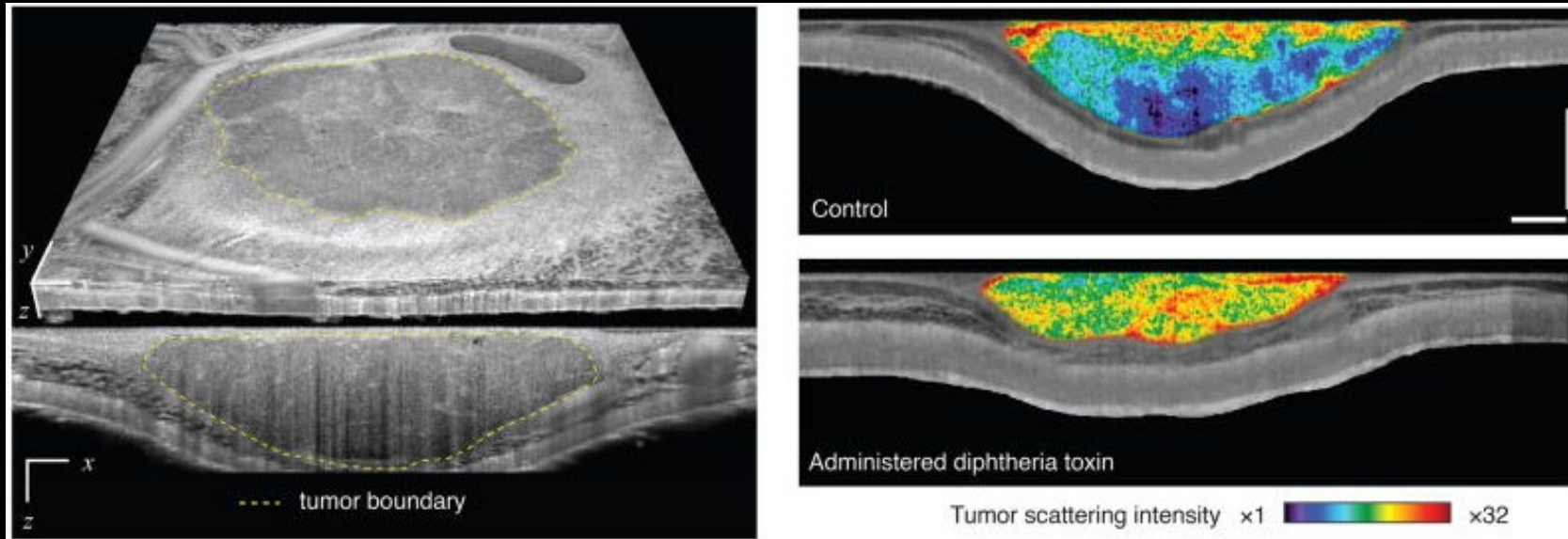


Retinal scanning and angiography to diagnose glaucoma, diabetic retinopathy, macular degeneration, etc.

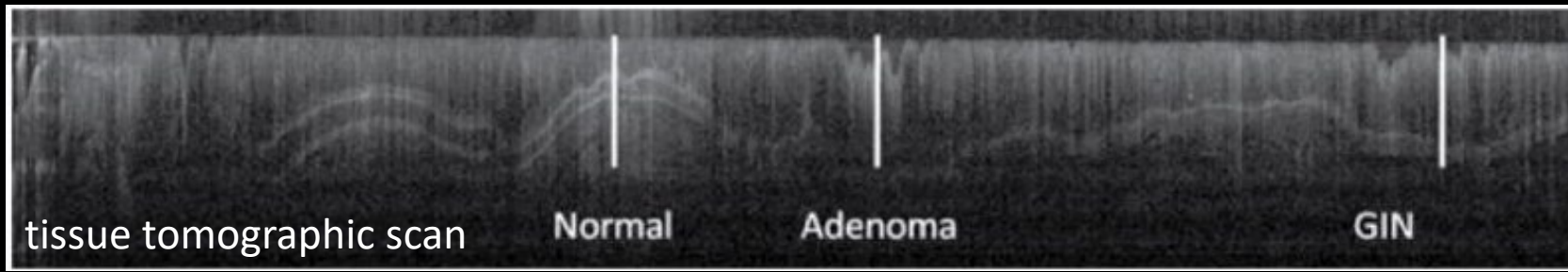
"Around 64% of adults working in computer and mathematical sciences use some vision correction option"

From the Washington Post, May 2023

Skin cancer imaging



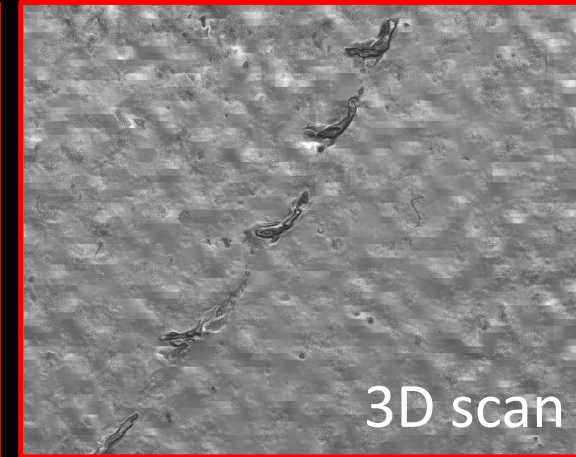
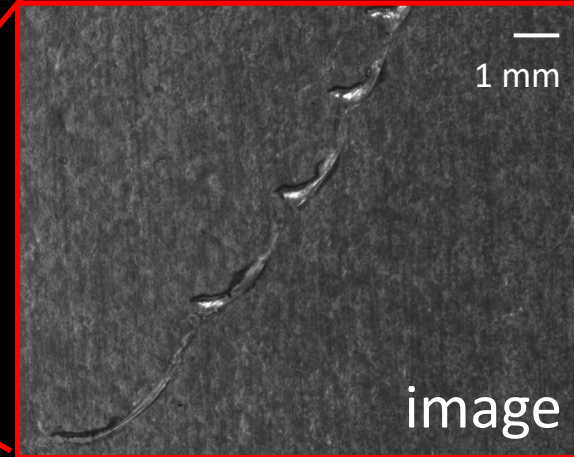
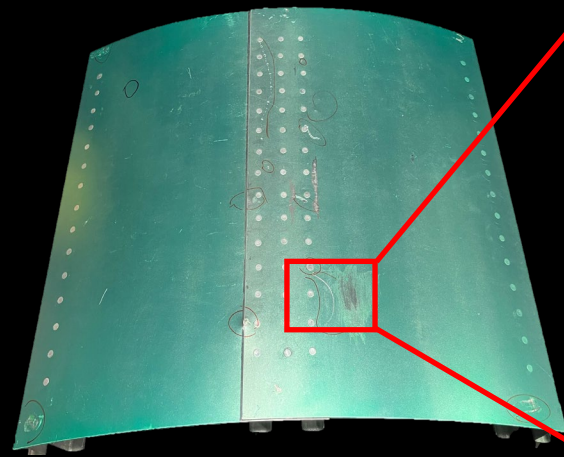
Non-invasive imaging of
breast and dorsal skin tumors
(Still at the pre-clinical level)



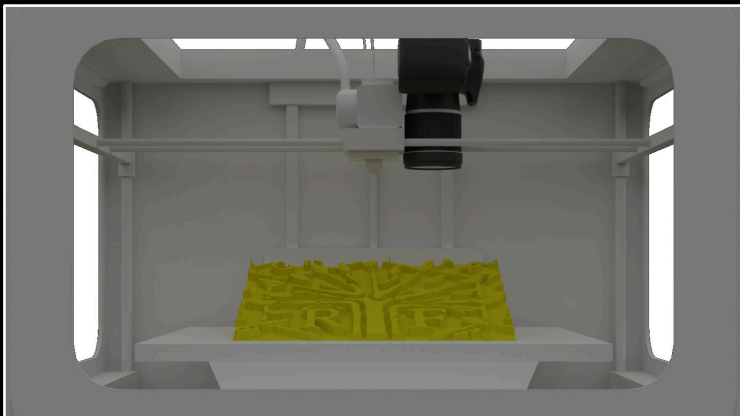
2 mm depth
10 μm resolution

Micrometer 3D sensing

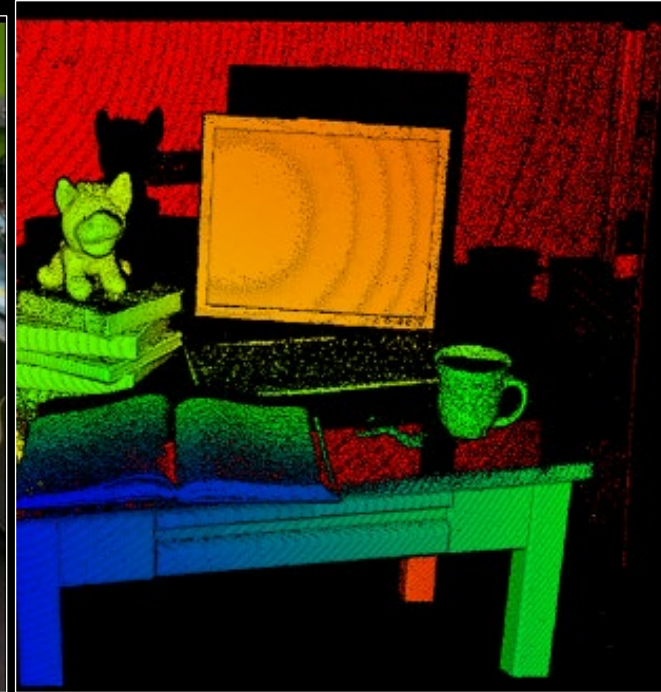
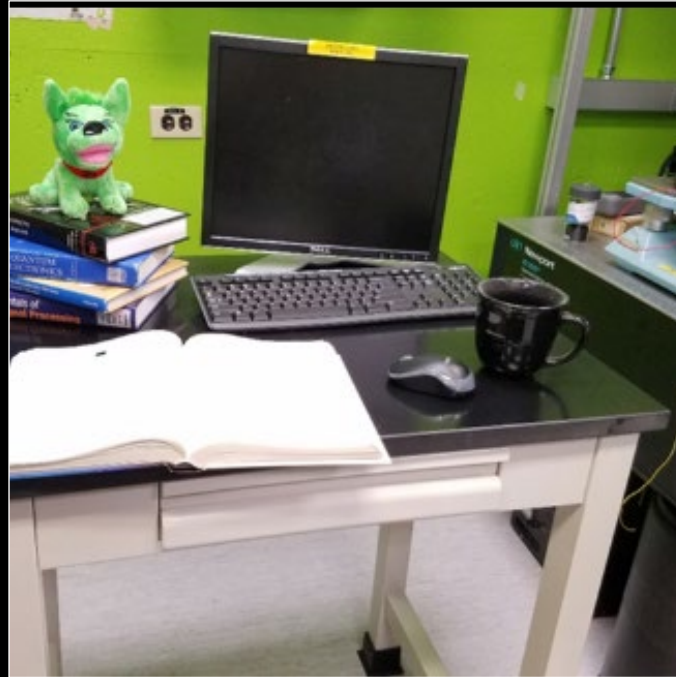
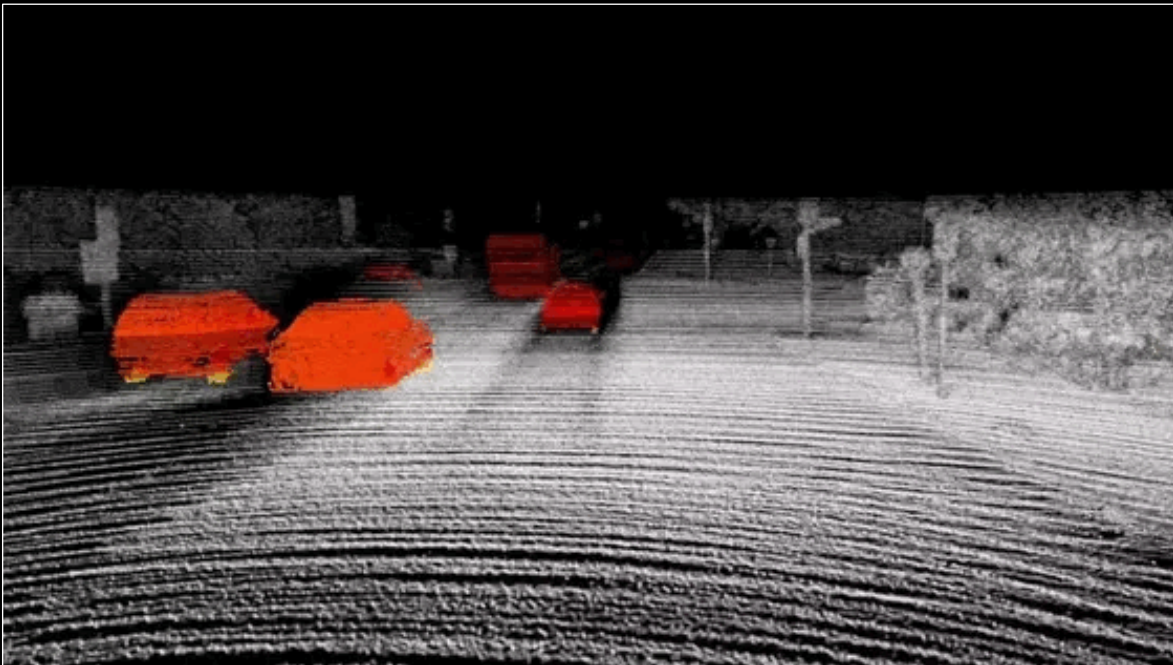
Inspection: aircraft fuselage section



Fabrication: 3D-printed Euro coin

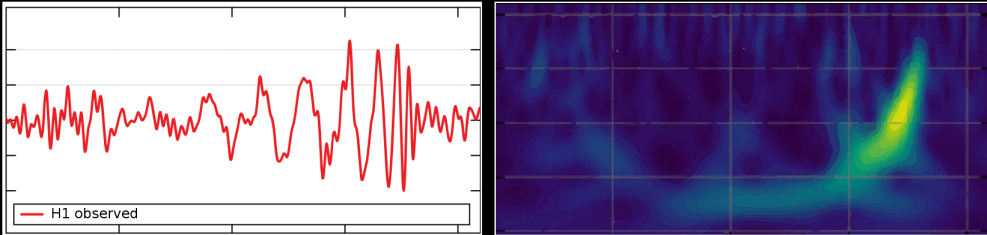


FMCW “4D” lidar

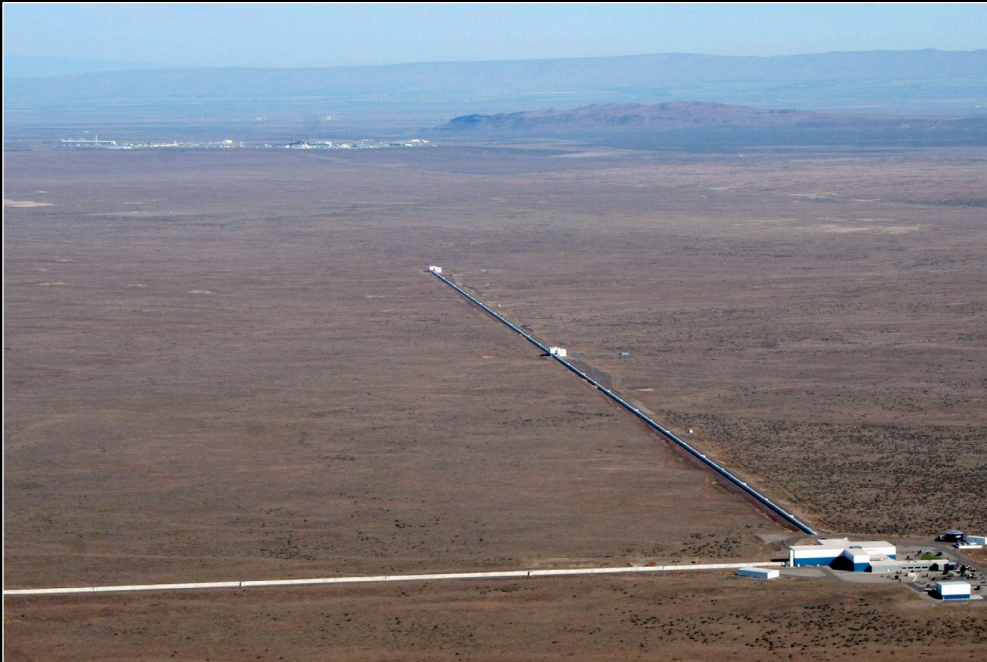


Depth *and* velocity using frequency-modulated continuous-wave lidar
(a.k.a. swept-source optical coherence tomography)

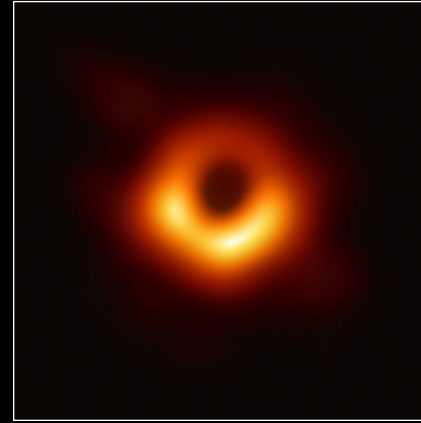
Very large-scale physics



first gravitational wave detection



Laser Interferometer Gravitational-wave Observatory (LIGO) at Hanford, WA
(4 km-long Michelson interferometer)

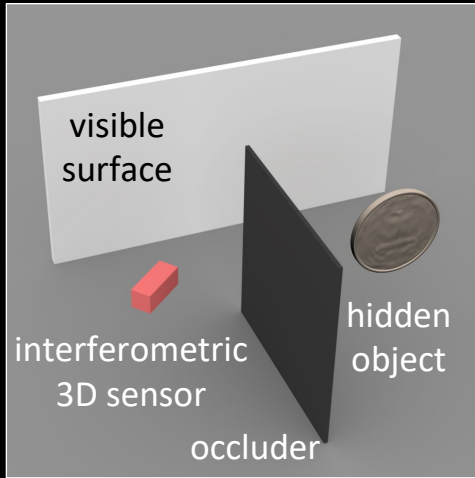


first image of a black hole
(center of Messier 87 galaxy)

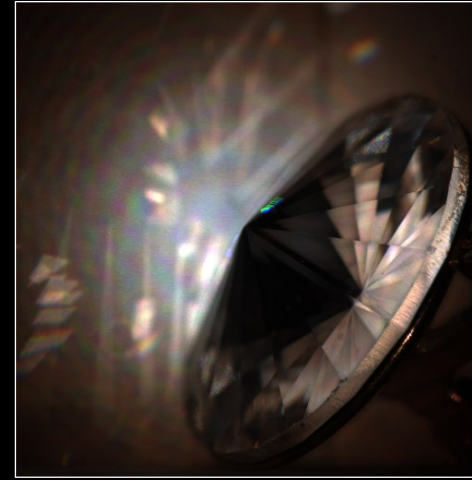


Location of observatories making up the
Event Horizon Telescope (EHT)
(very-long-baseline interferometry)

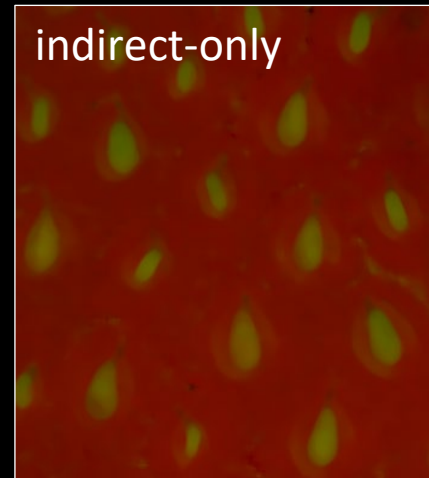
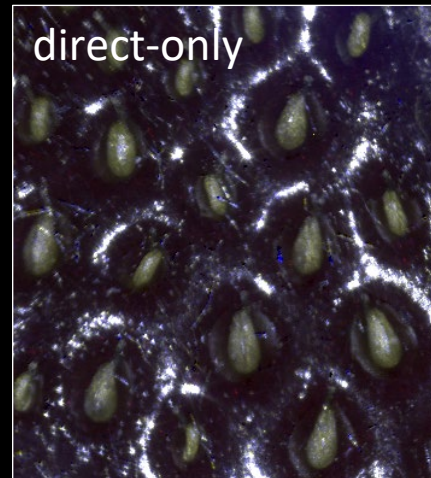
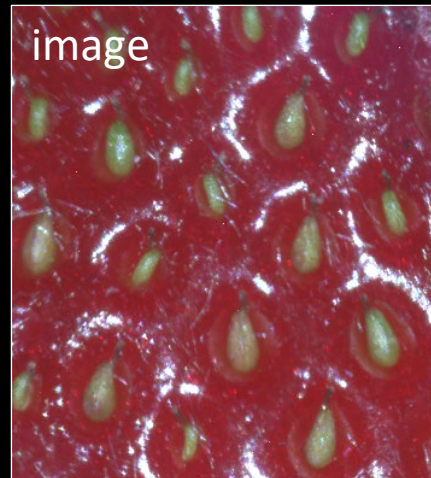
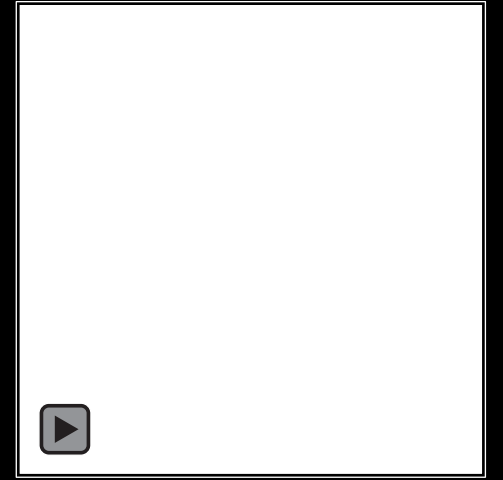
Computational imaging



non-line-of-sight imaging



transient imaging



separation of direct-indirect illumination

We will focus on micron-scale 3D scanning

You will build a setup that can produce scans such as these

captured image

coin



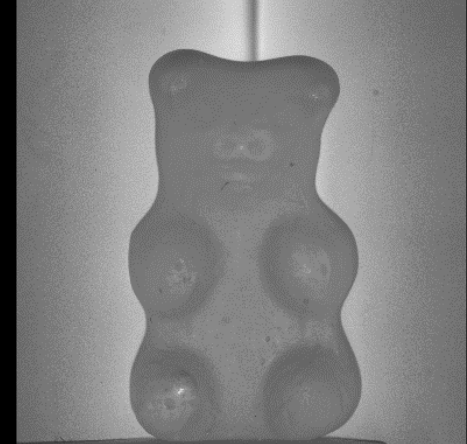
gnocchi



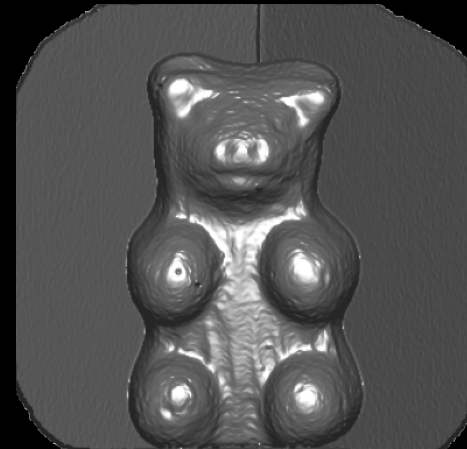
soap carving



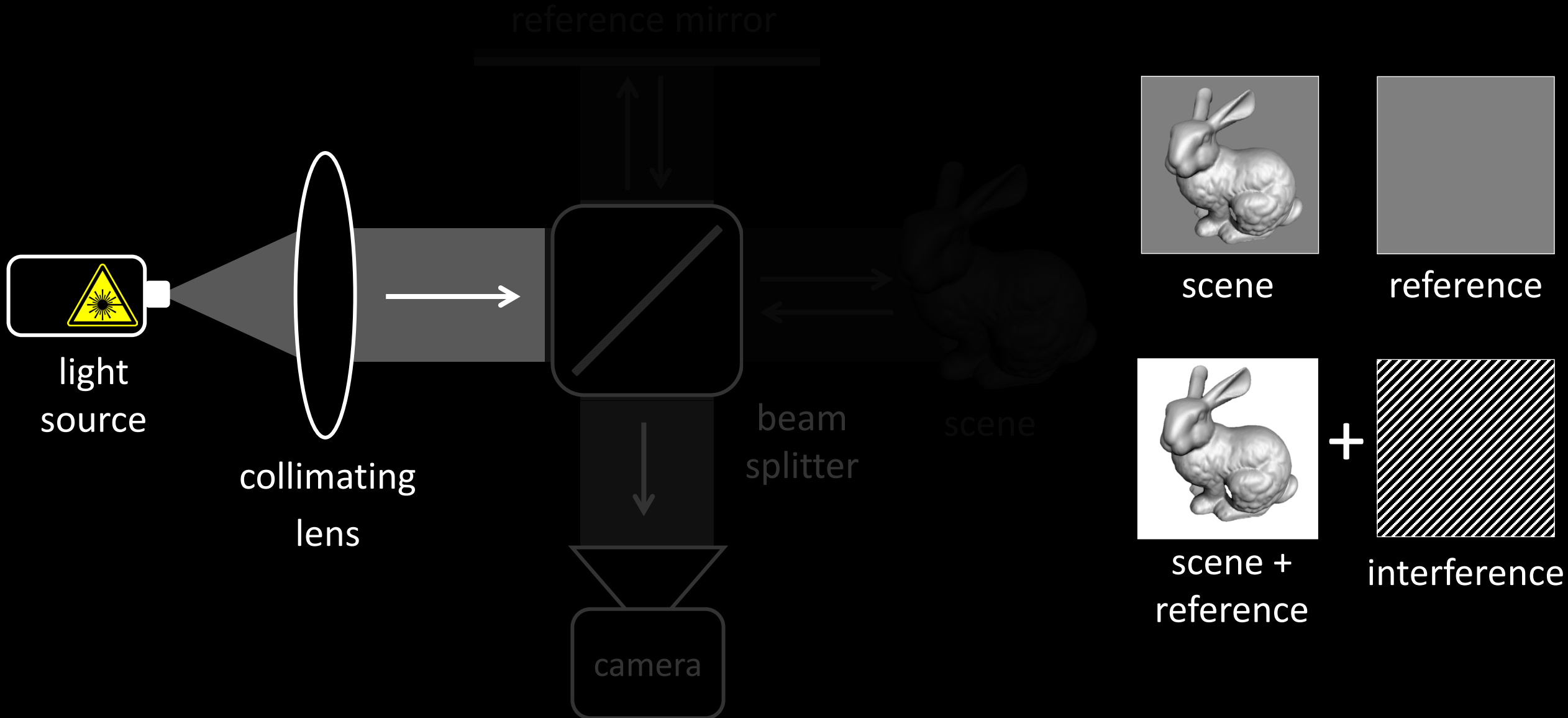
gummy bear



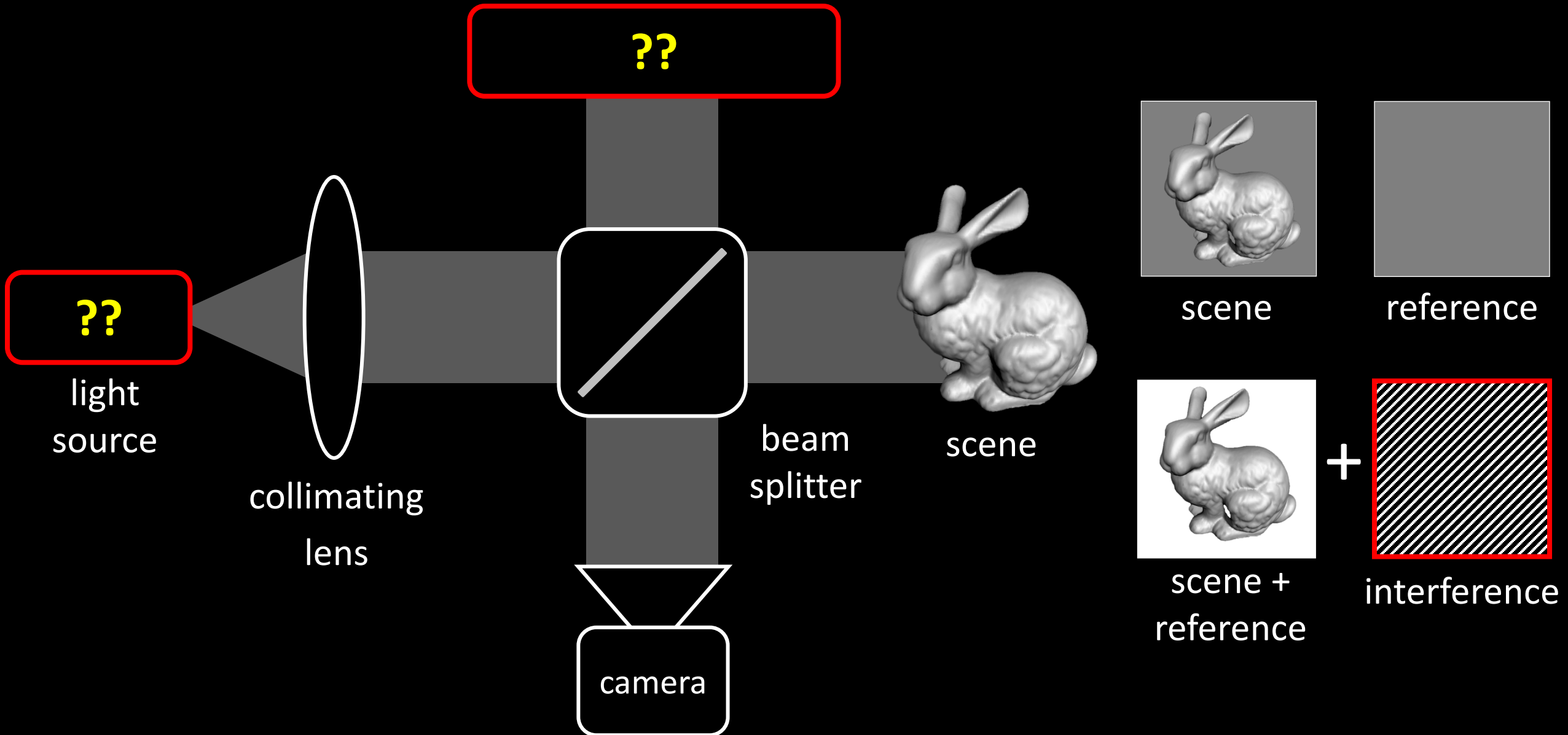
scanned shape



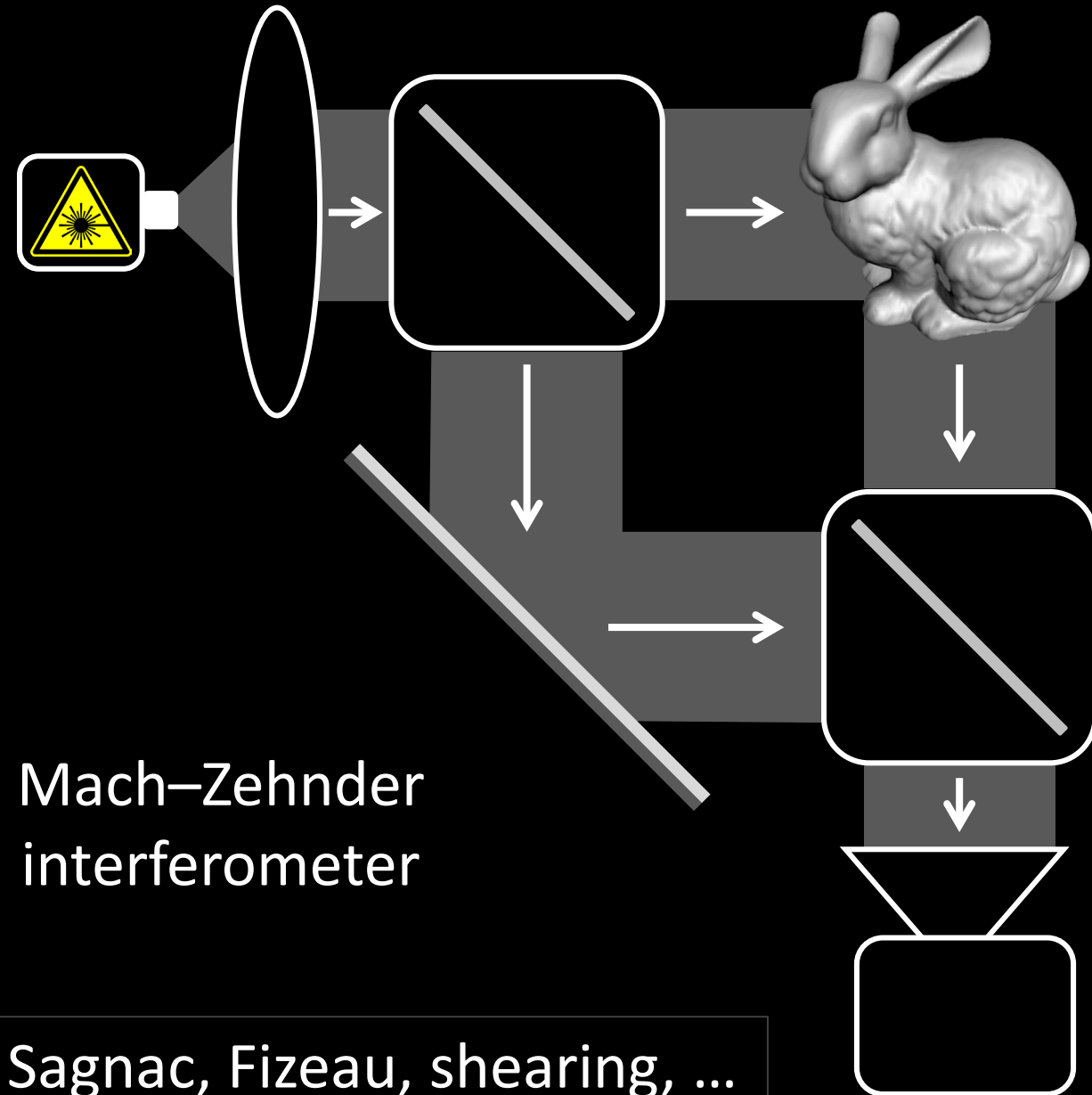
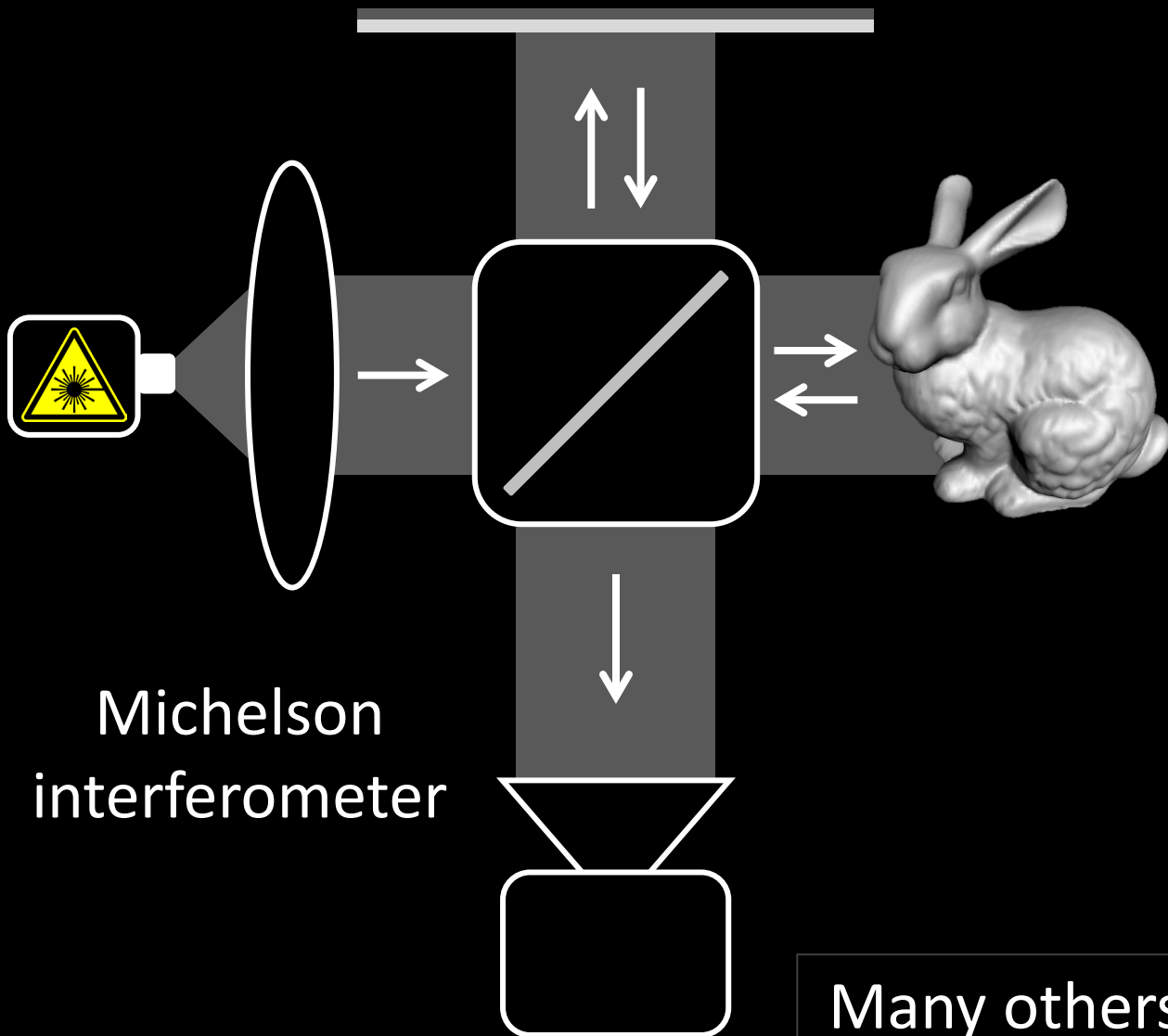
Interferometric imaging



Interferometric imaging

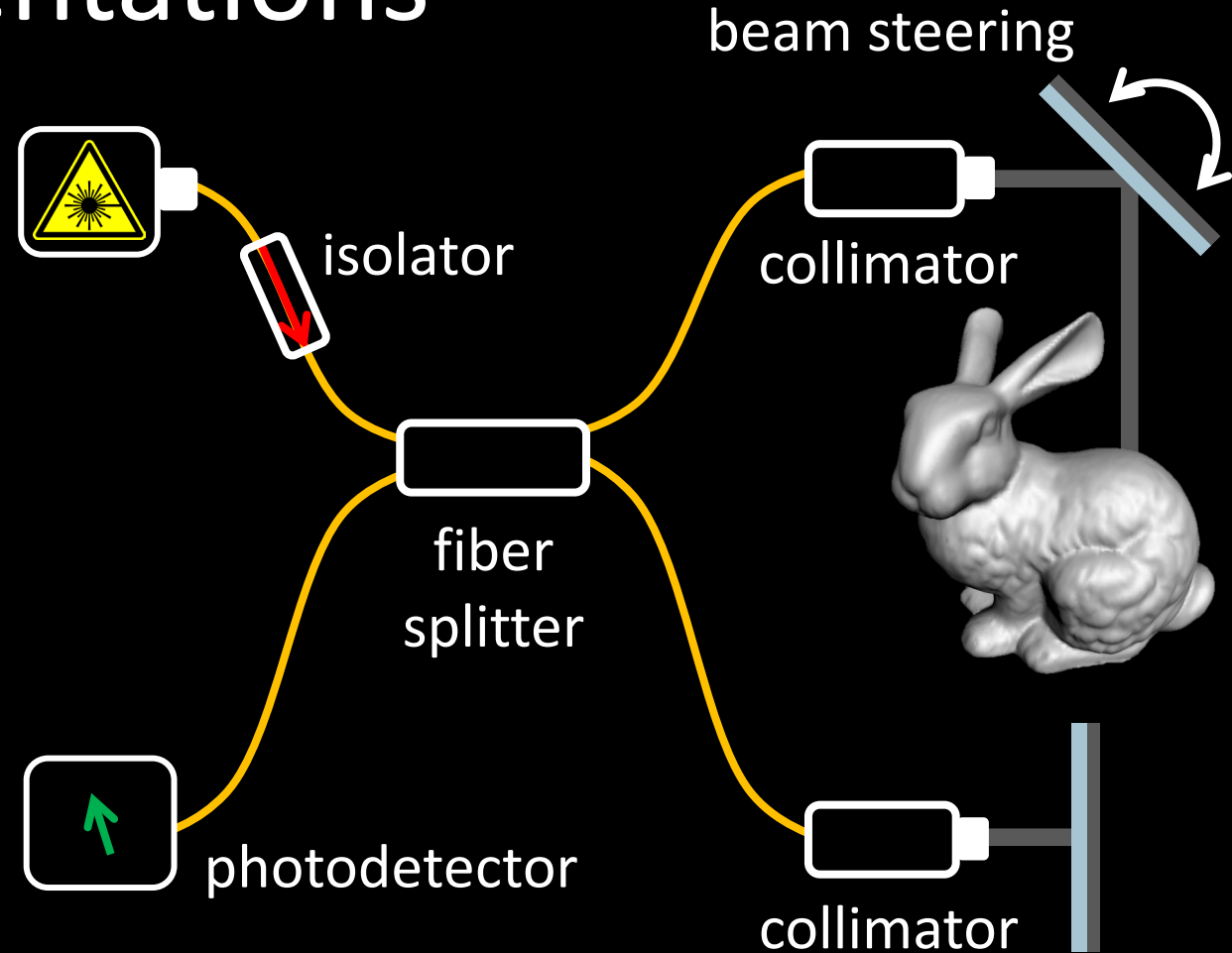
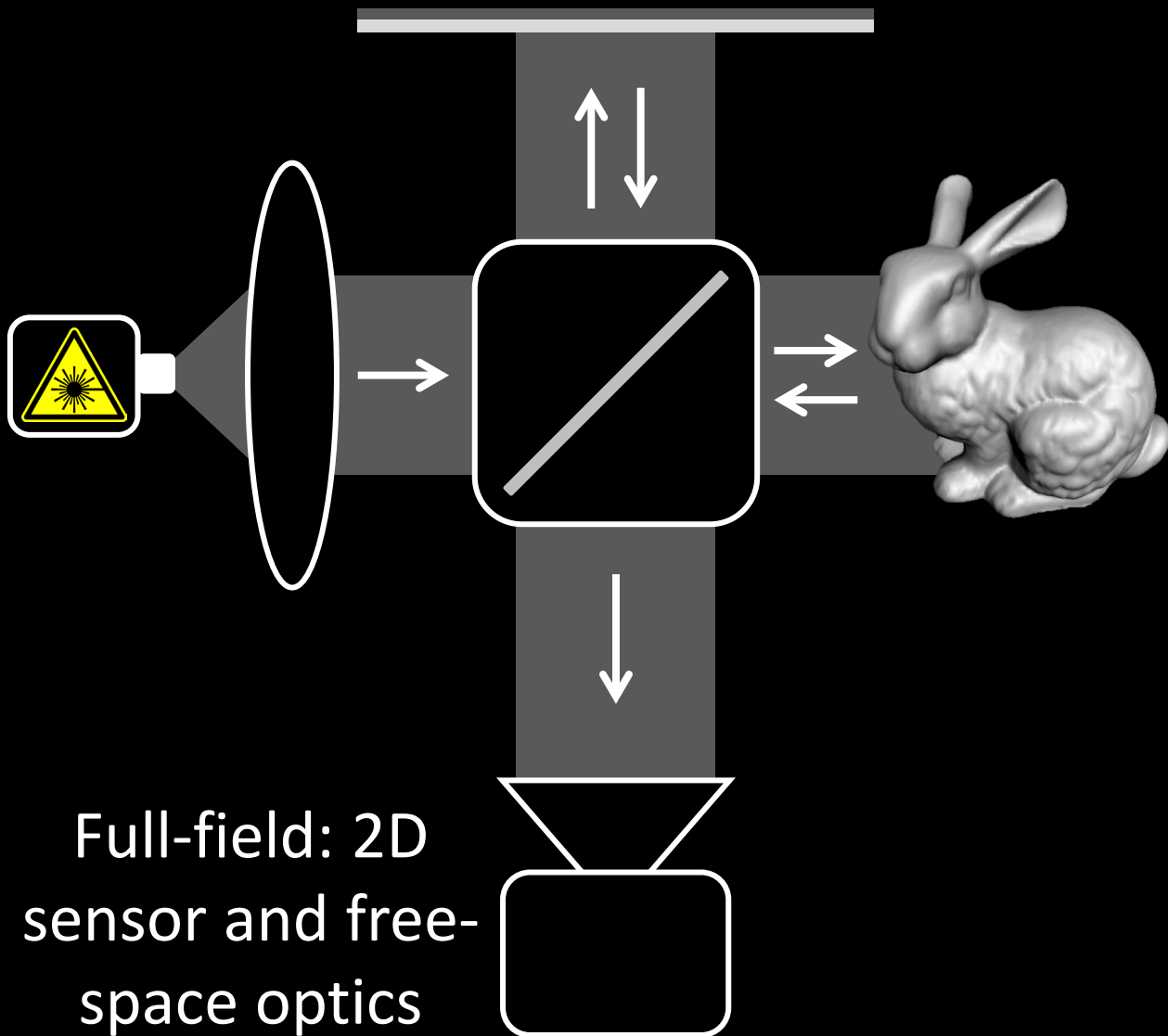


Interferometer designs

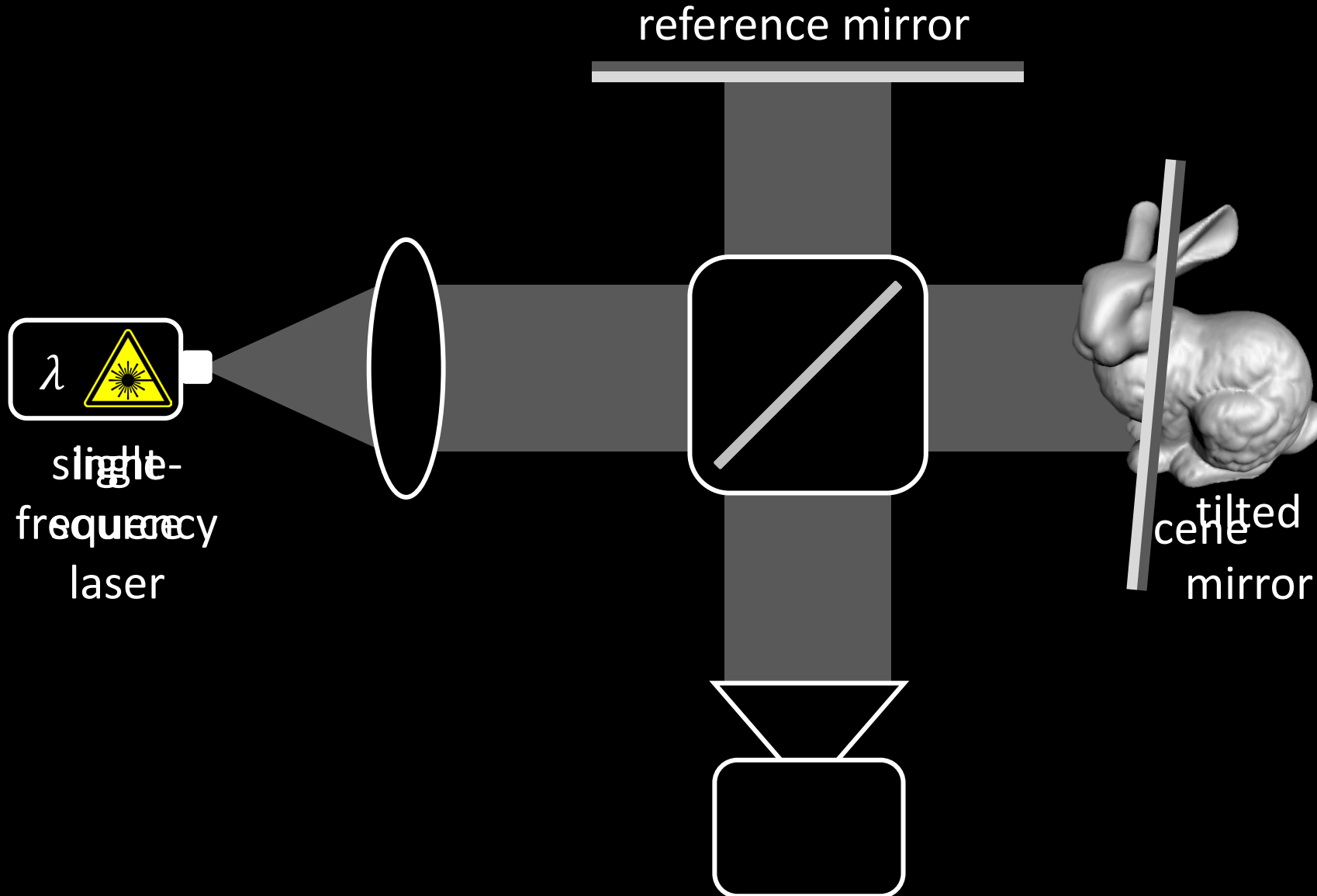


Many others: Sagnac, Fizeau, shearing, ...

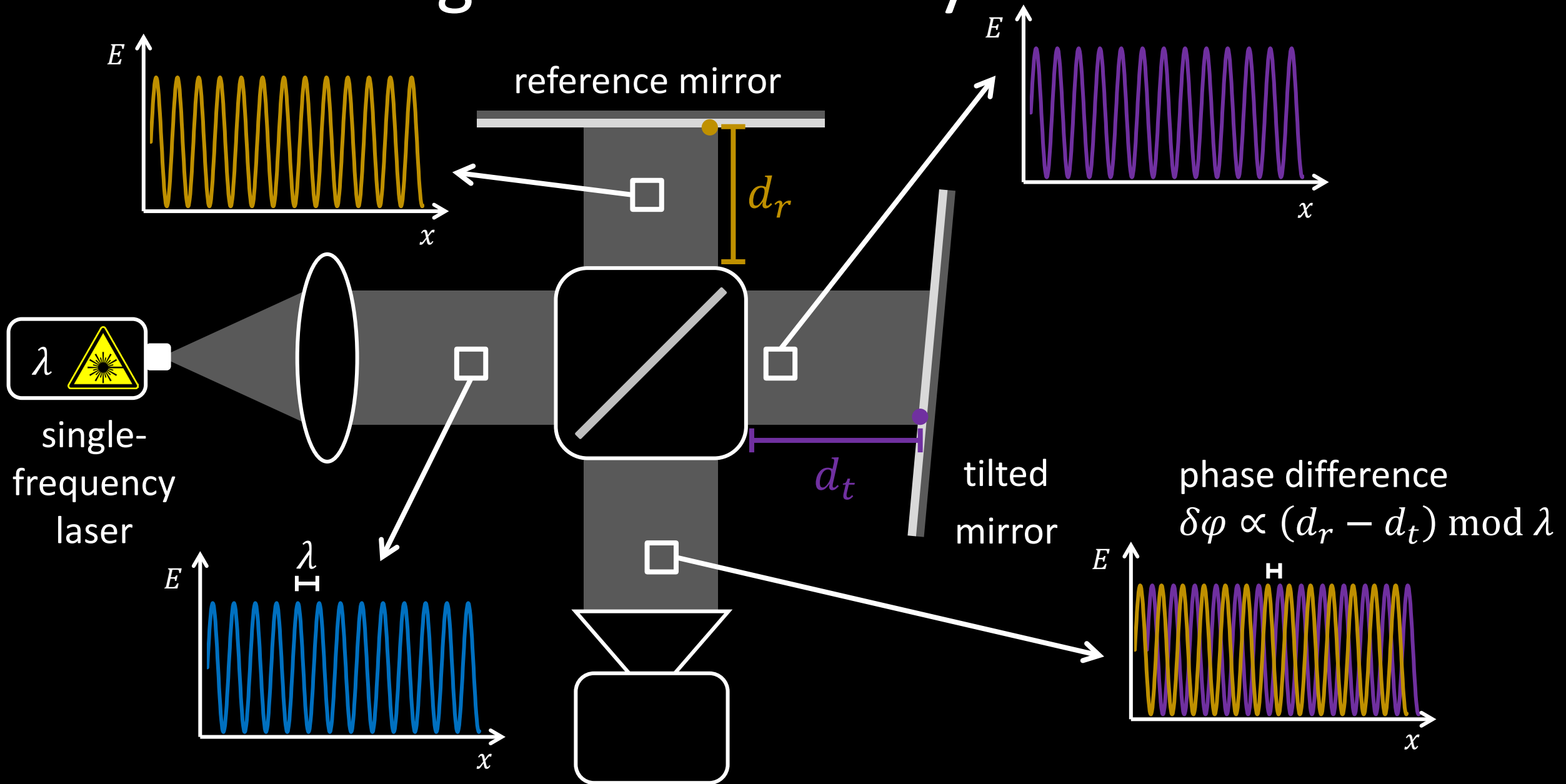
Interferometer implementations



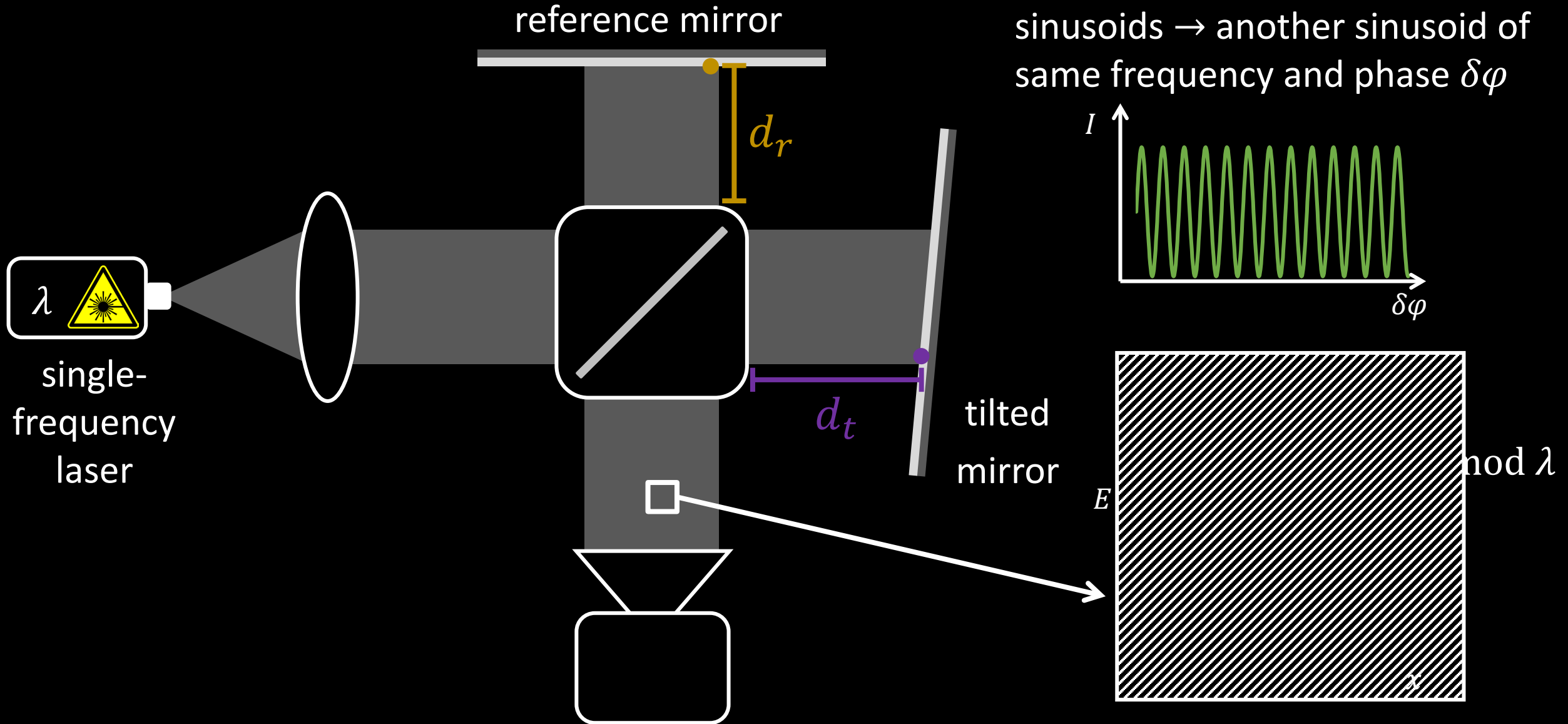
Phase-shifting interferometry



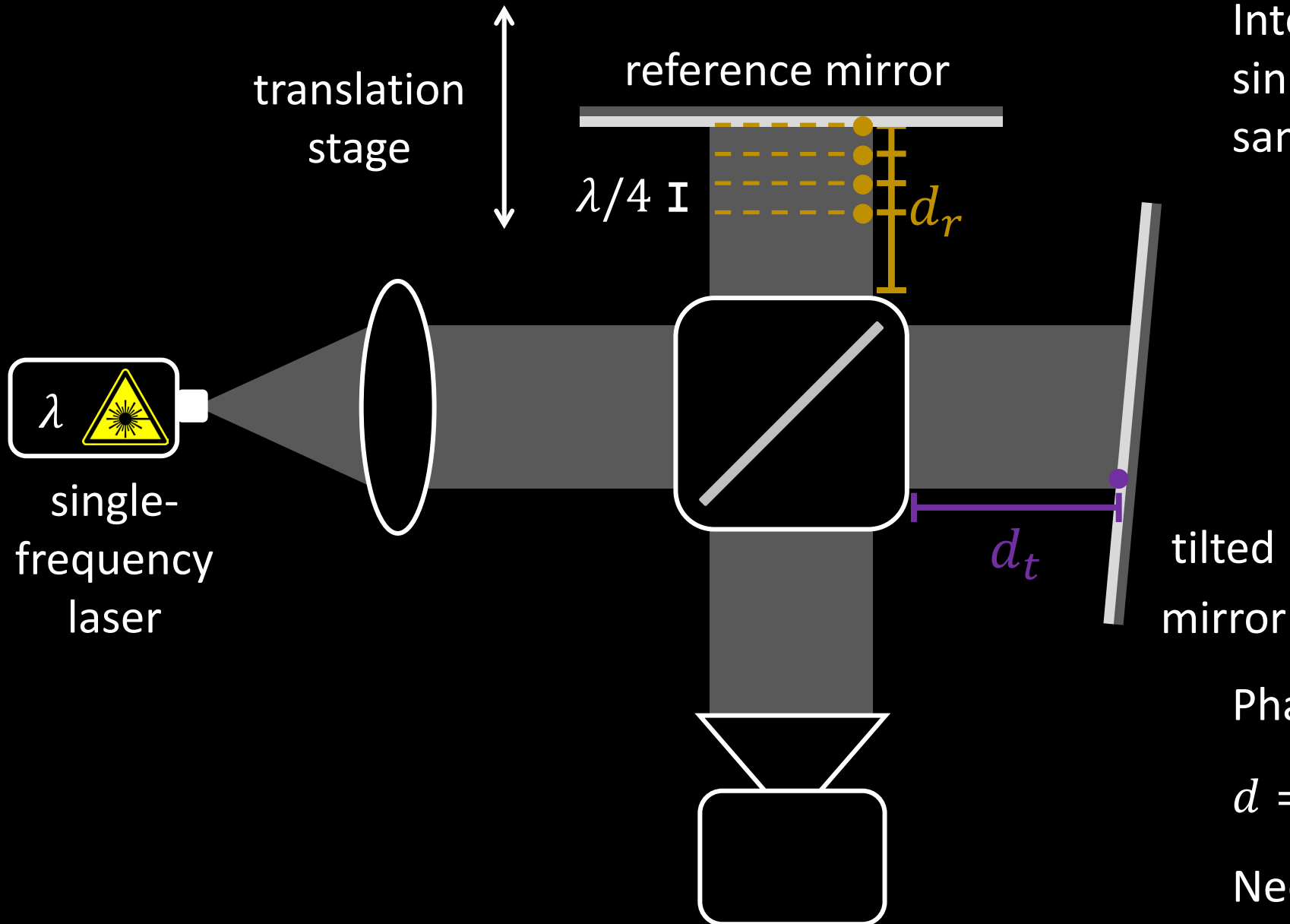
Phase-shifting interferometry



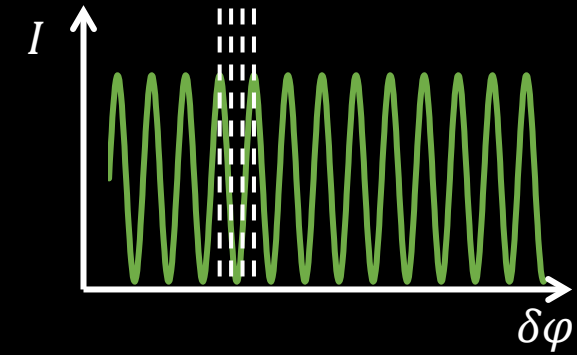
Phase-shifting interferometry



Phase-shifting interferometry



Interference: correlation of two sinusoids $\rightarrow$ another sinusoid of same frequency and phase $\delta\varphi$

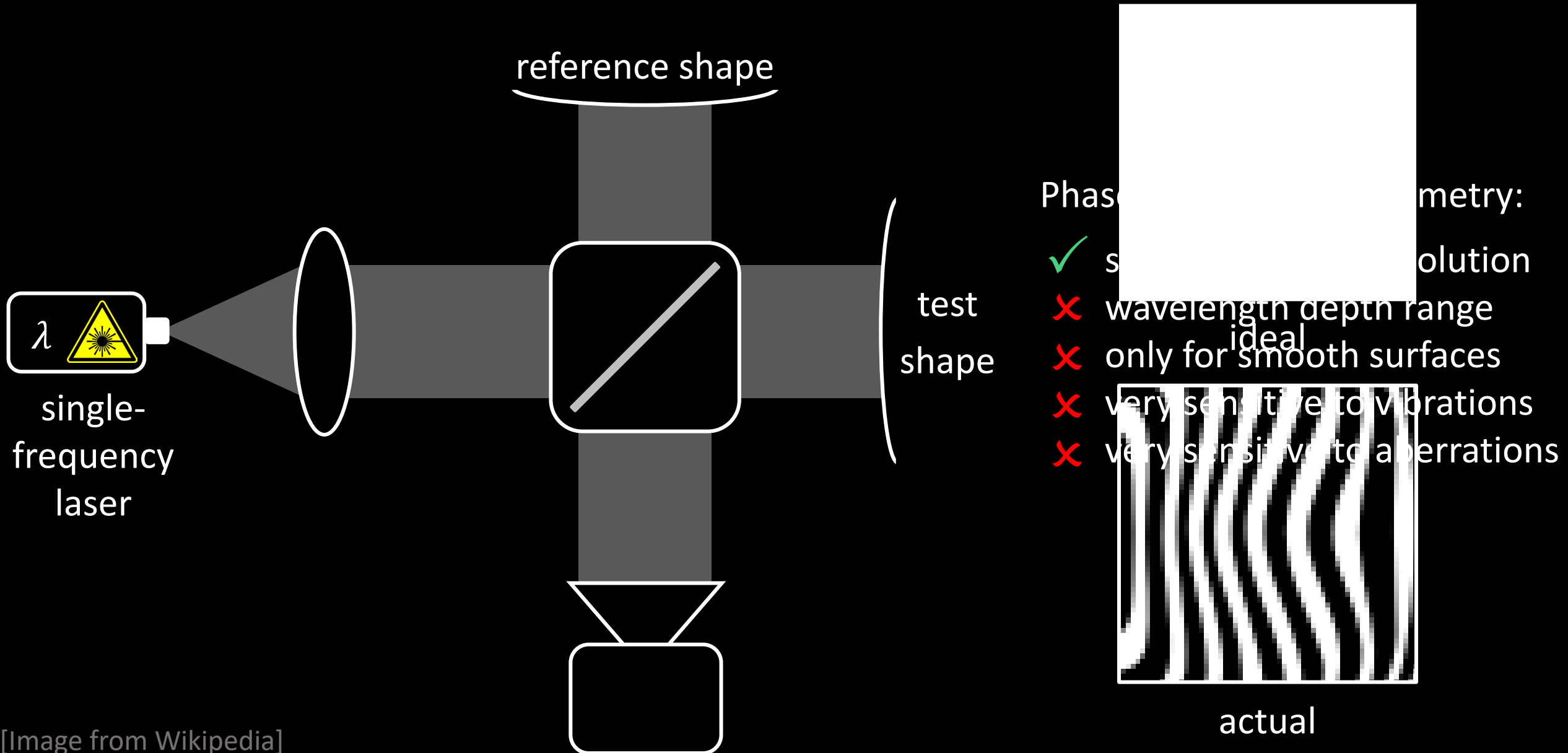


Phase retrieval:

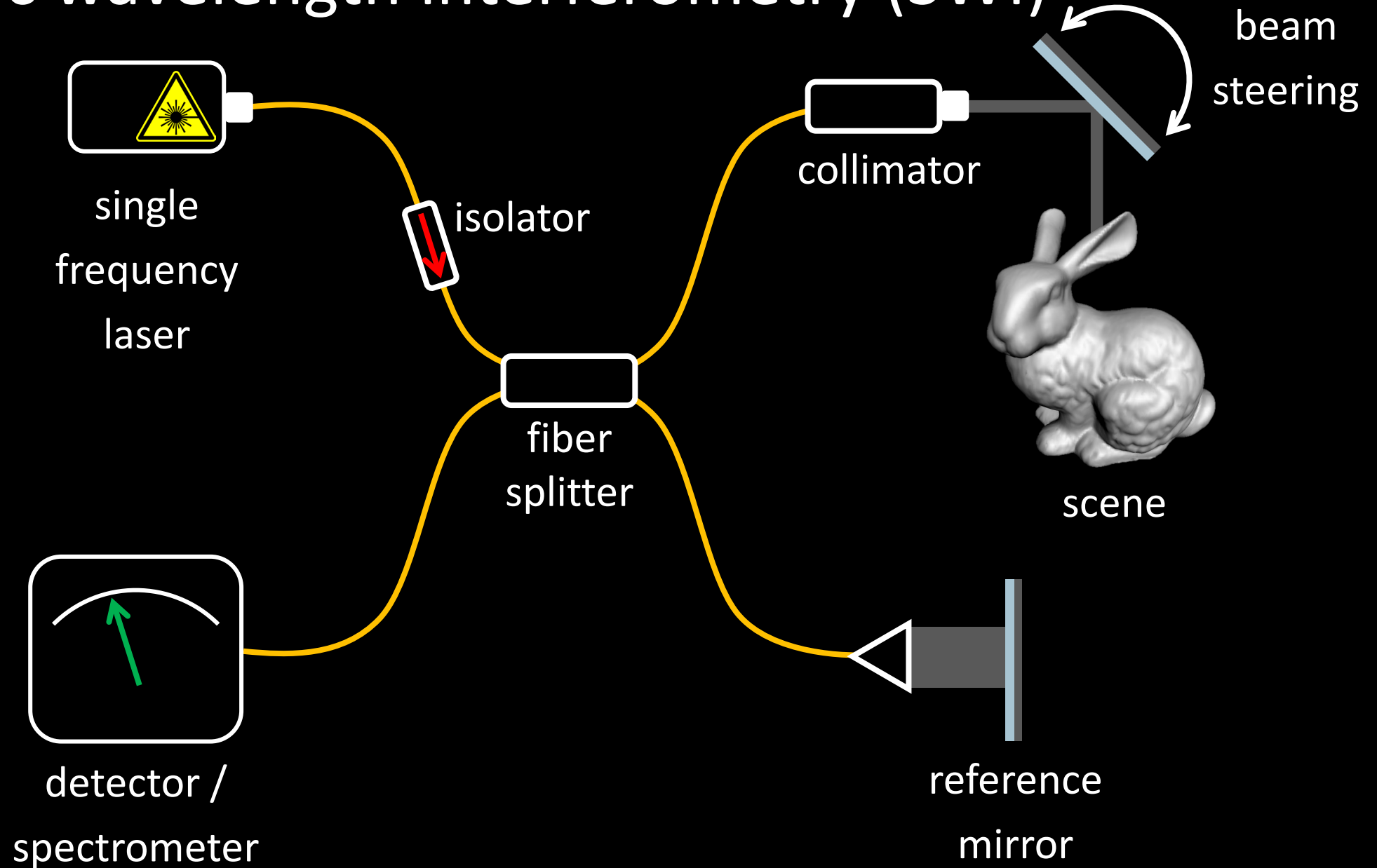
$$d = \frac{\lambda}{2\pi} \tan^{-1} \frac{E_4 - E_2}{E_1 - E_3} + n\lambda$$

Needs only 4 axial measurements

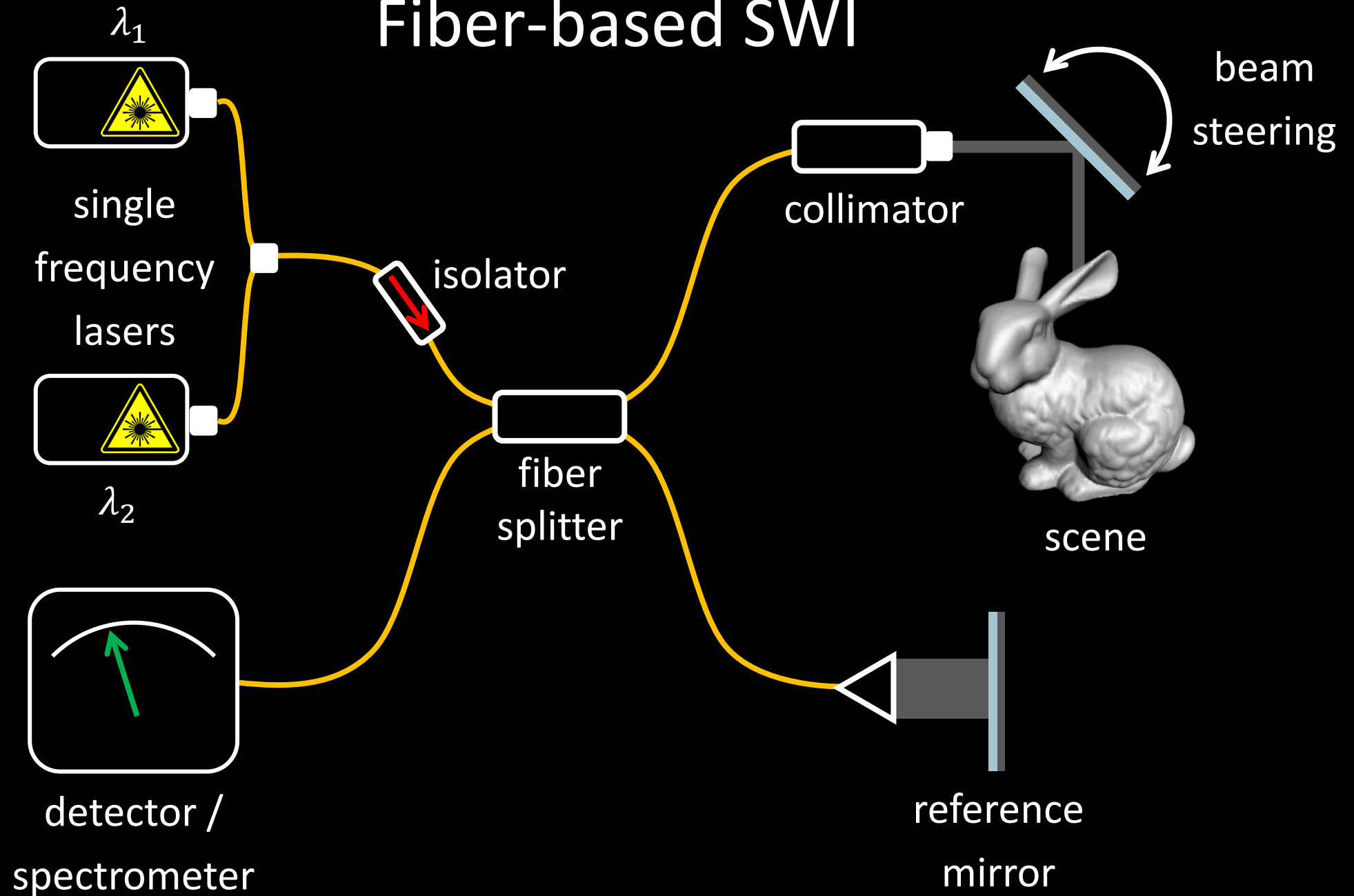
Example: surface deflectometry with PSI



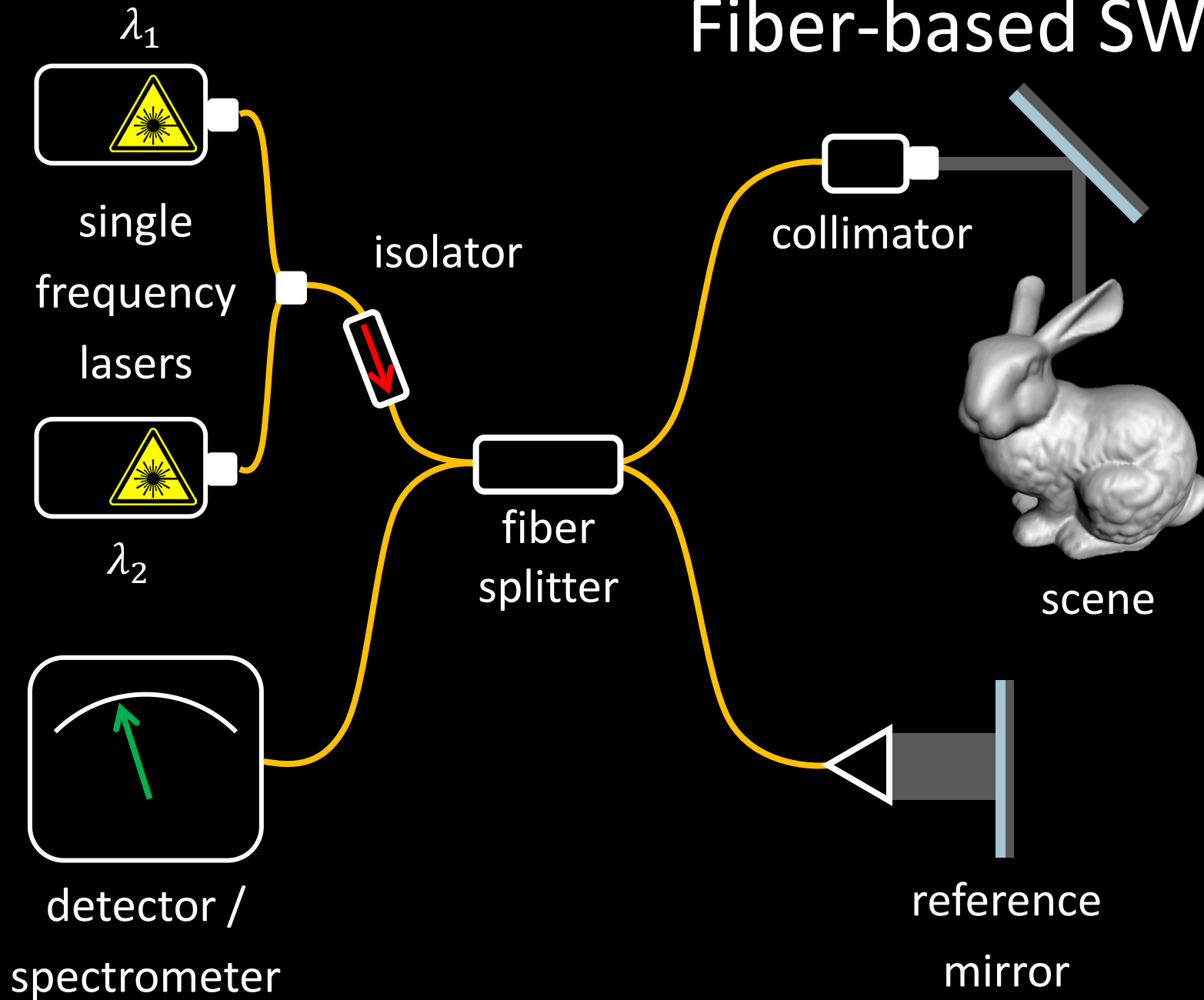
Synthetic wavelength interferometry (SWI)



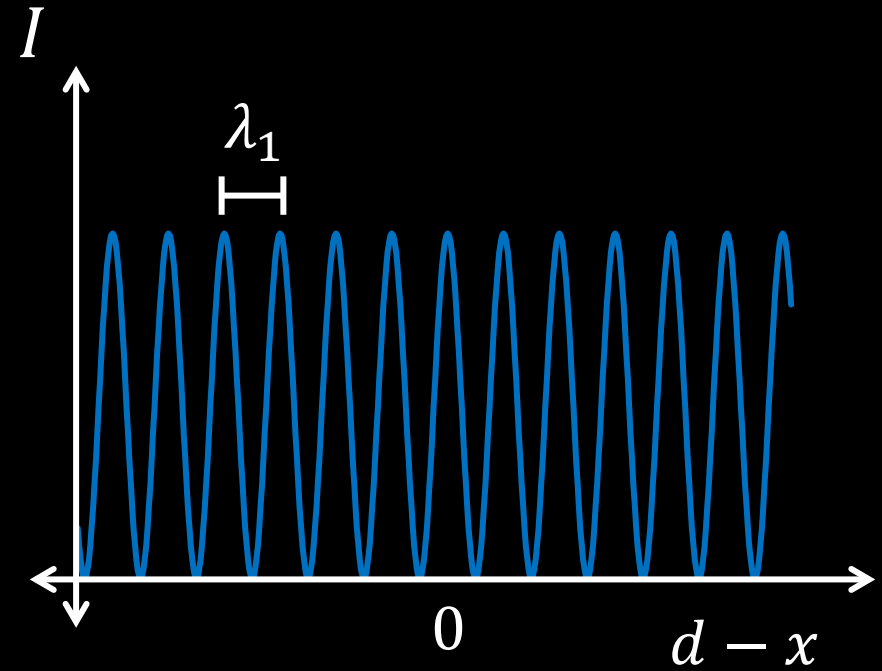
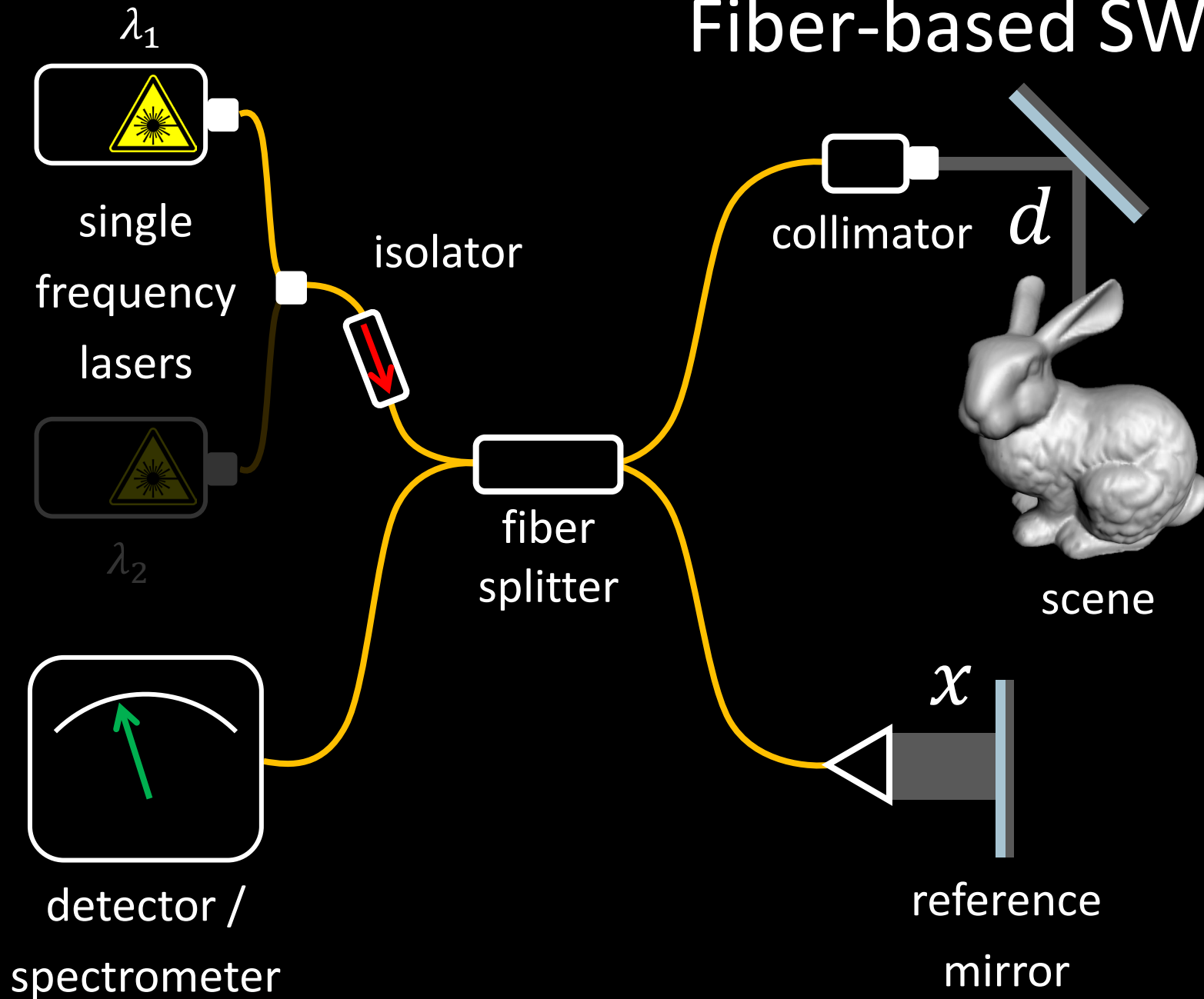
Fiber-based SWI



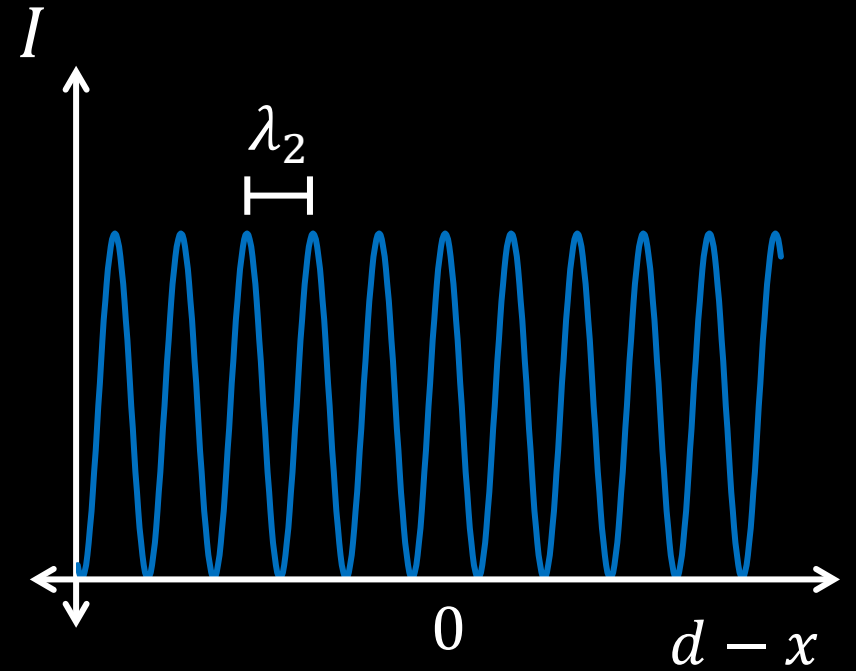
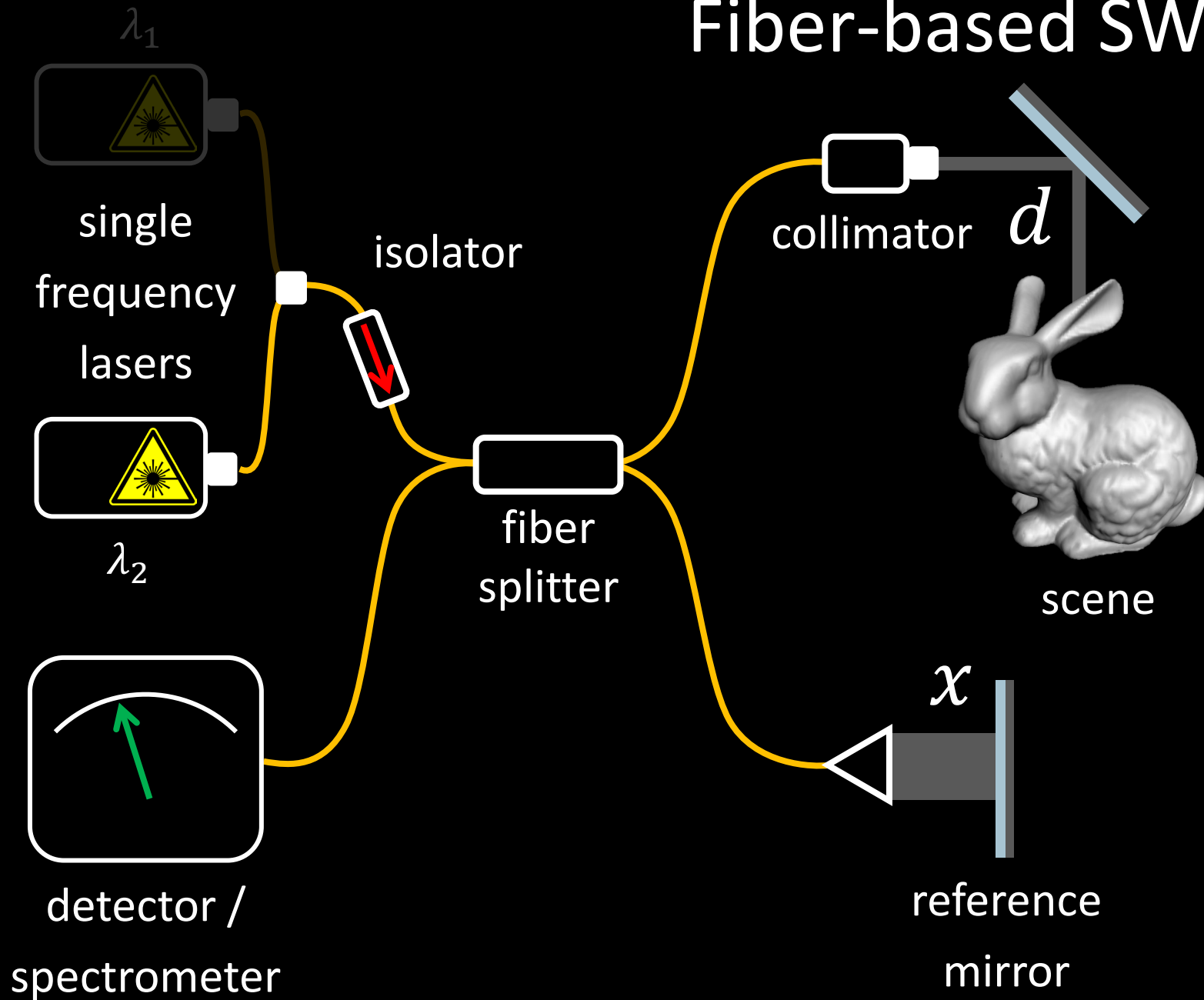
Fiber-based SWI



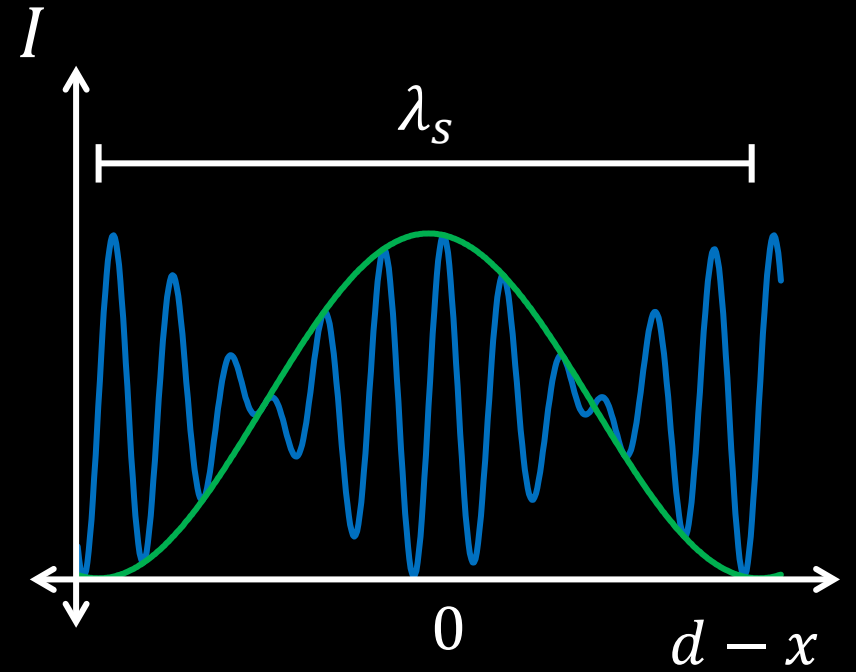
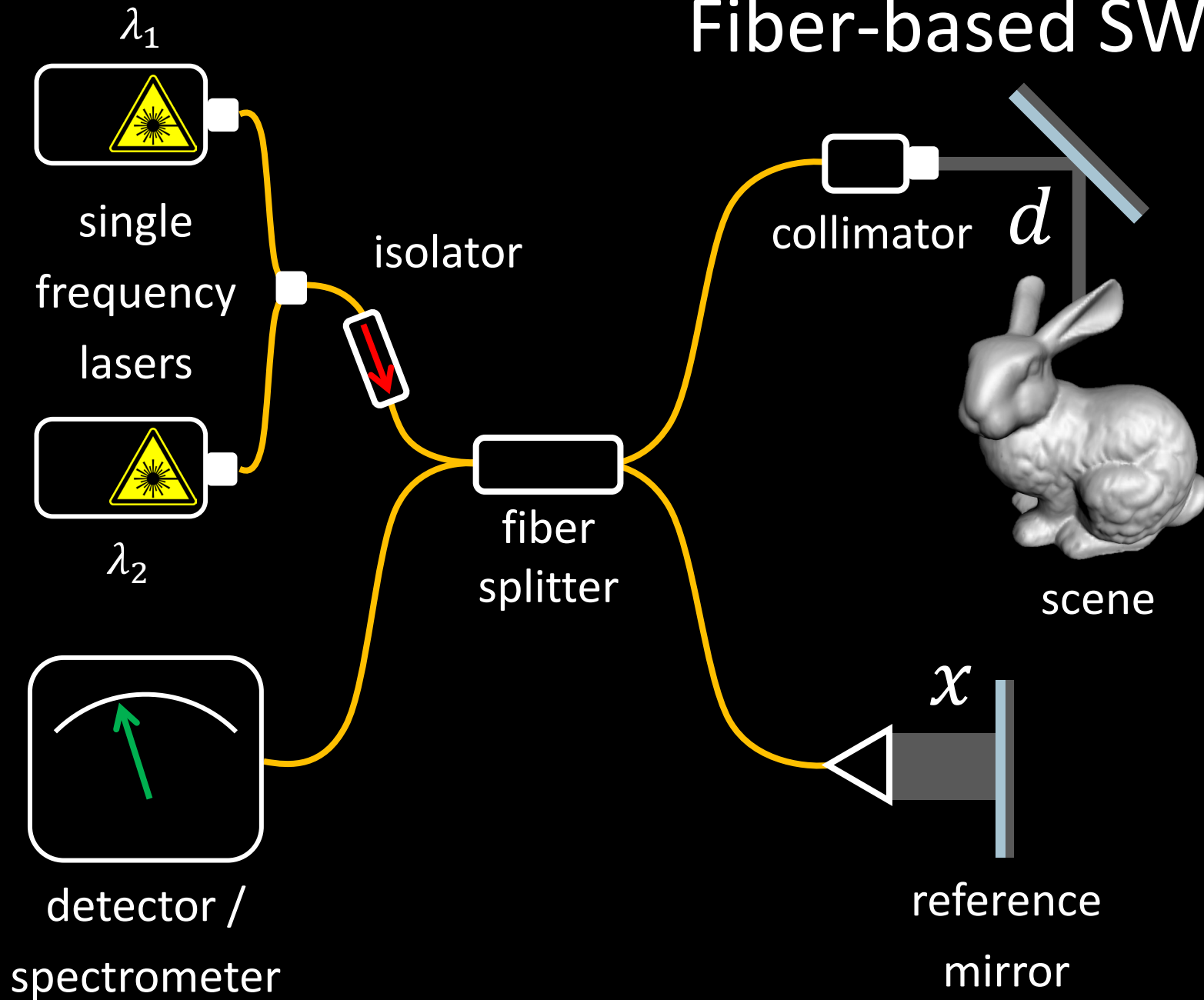
Fiber-based SWI



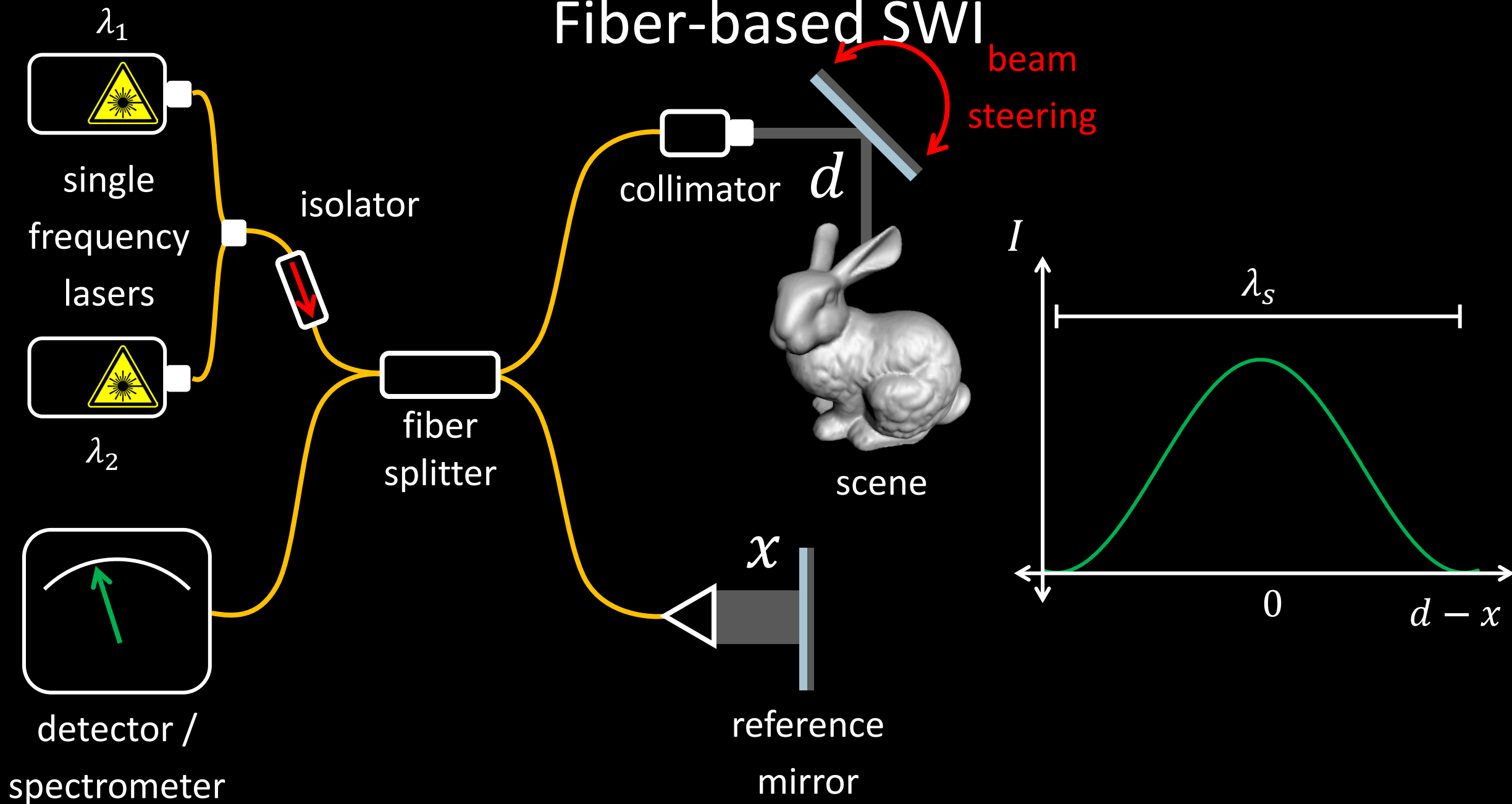
Fiber-based SWI



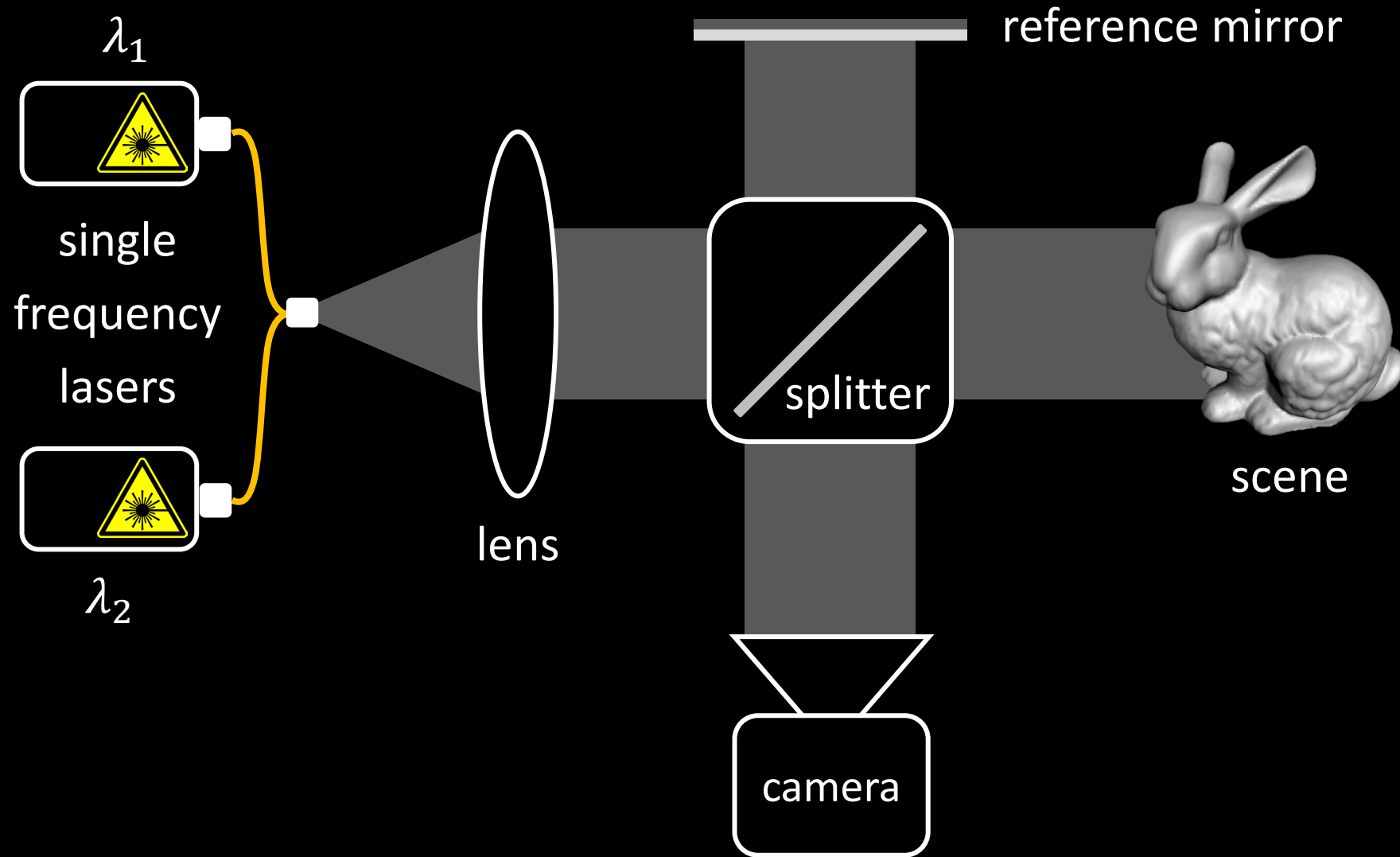
Fiber-based SWI



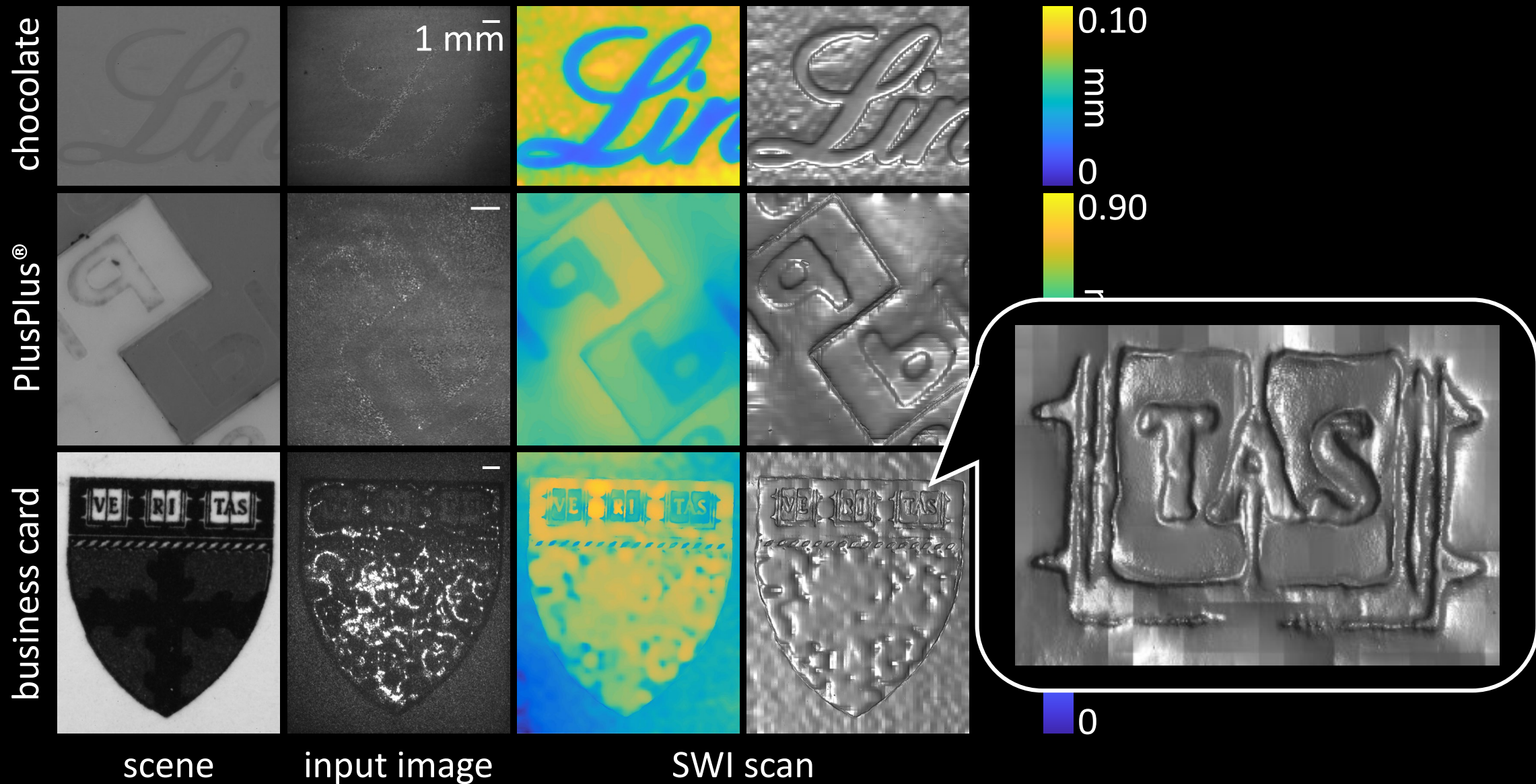
Fiber-based SWI



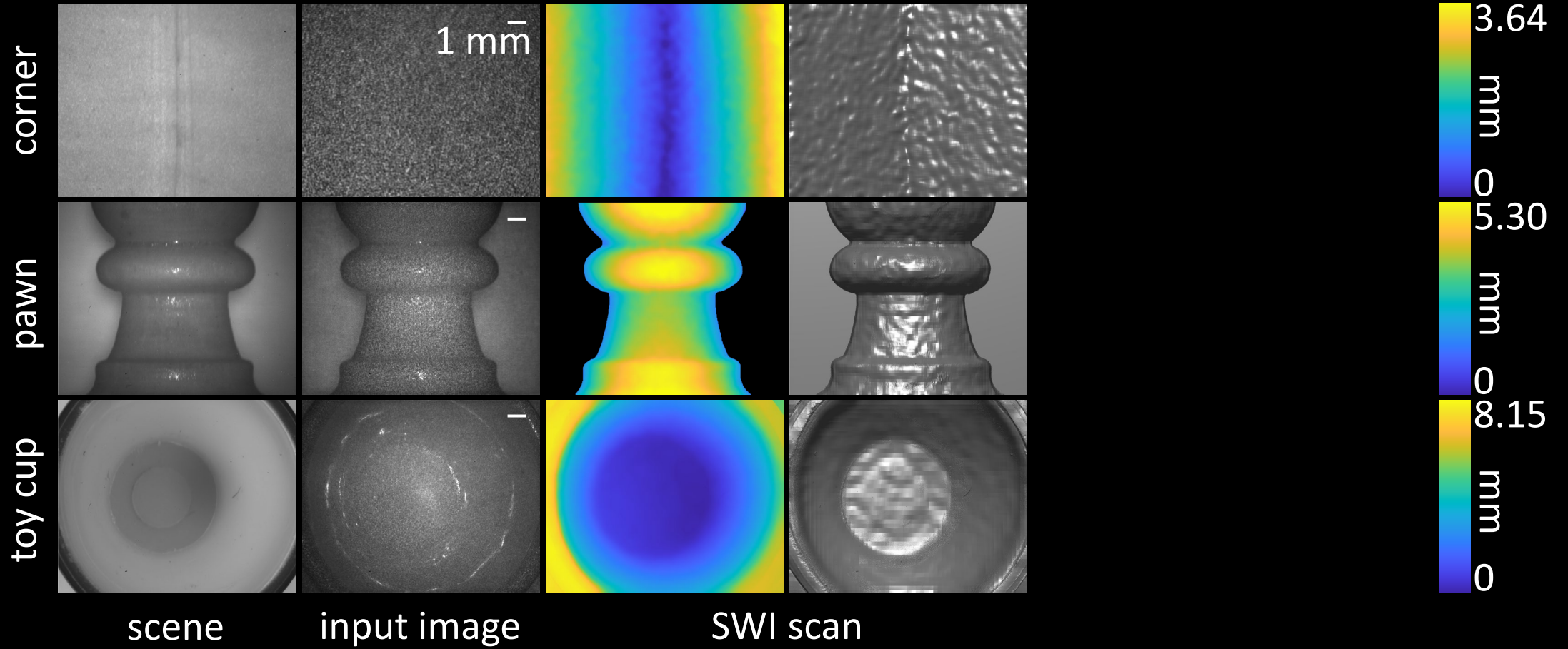
Full-field SWI



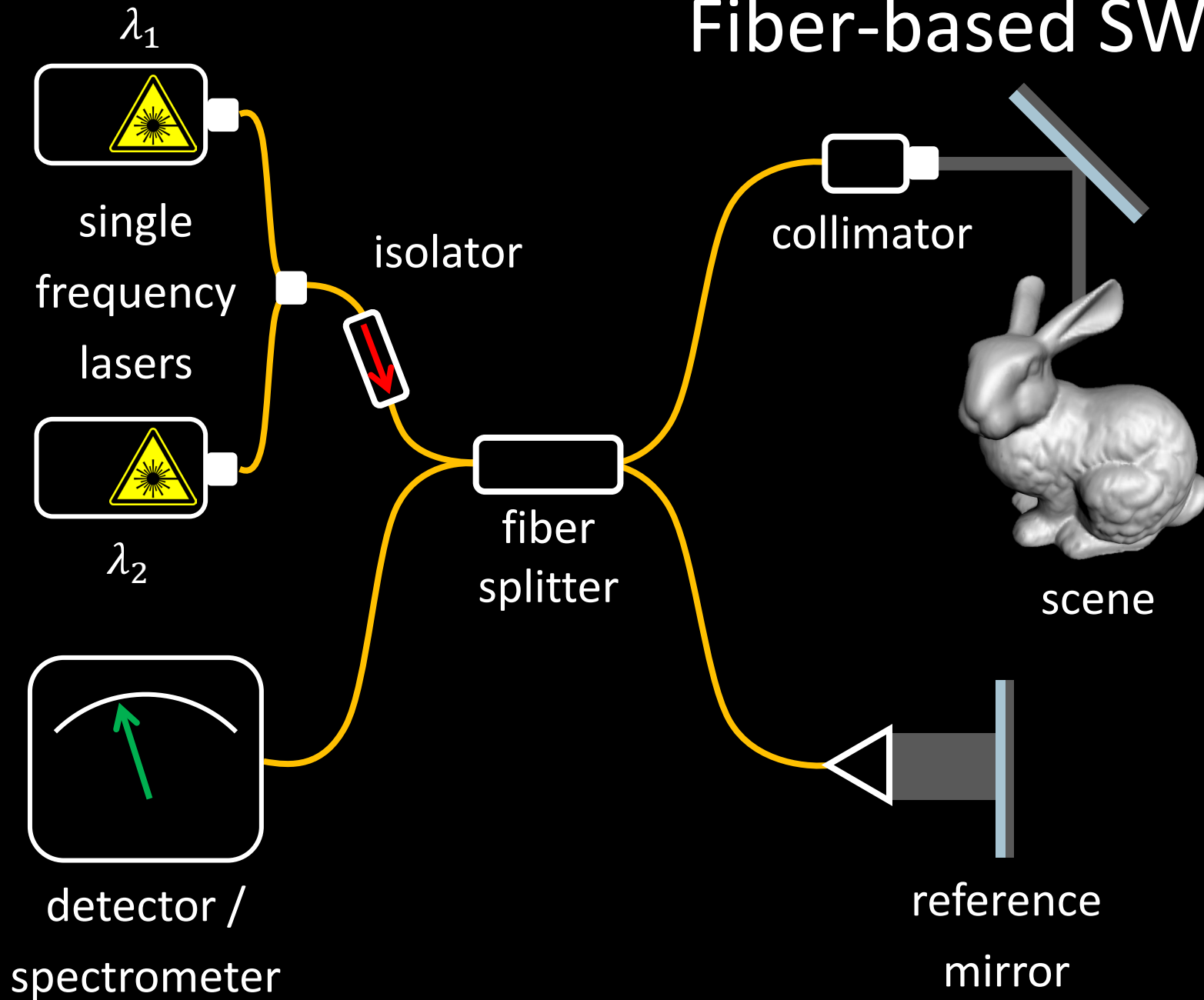
3D scanning at 1 mm depth range



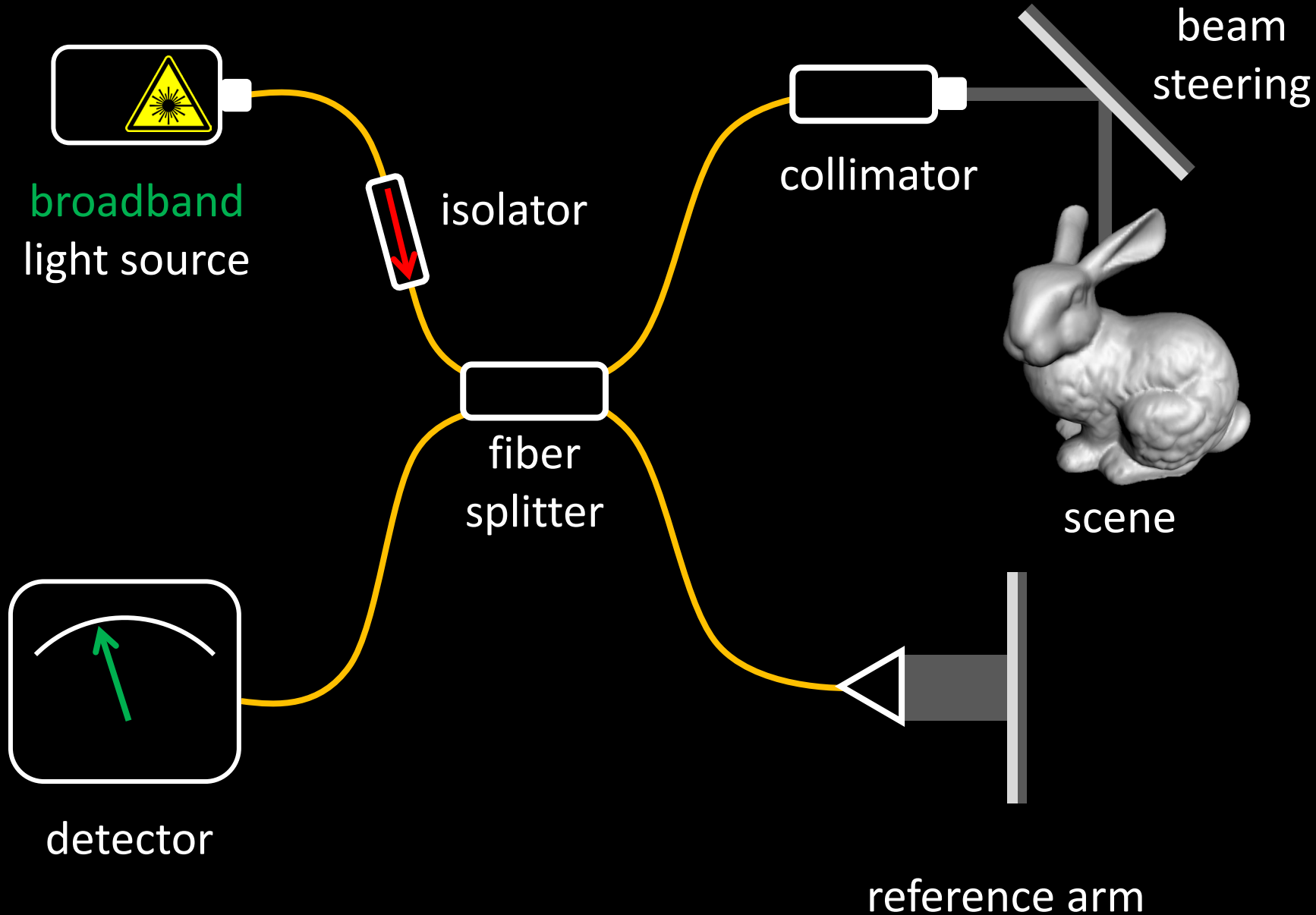
3D scanning at 1 cm depth range



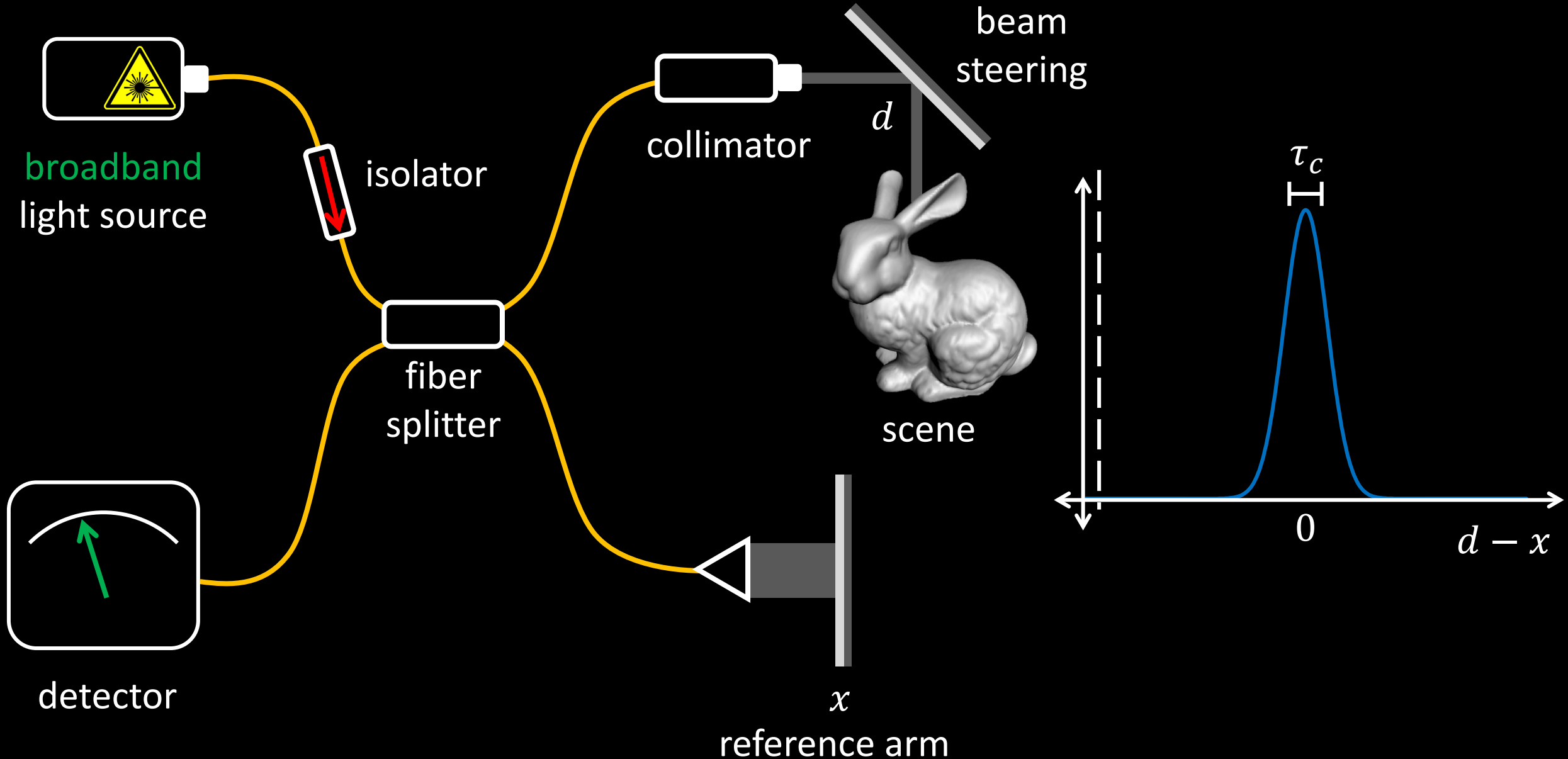
Fiber-based SWI



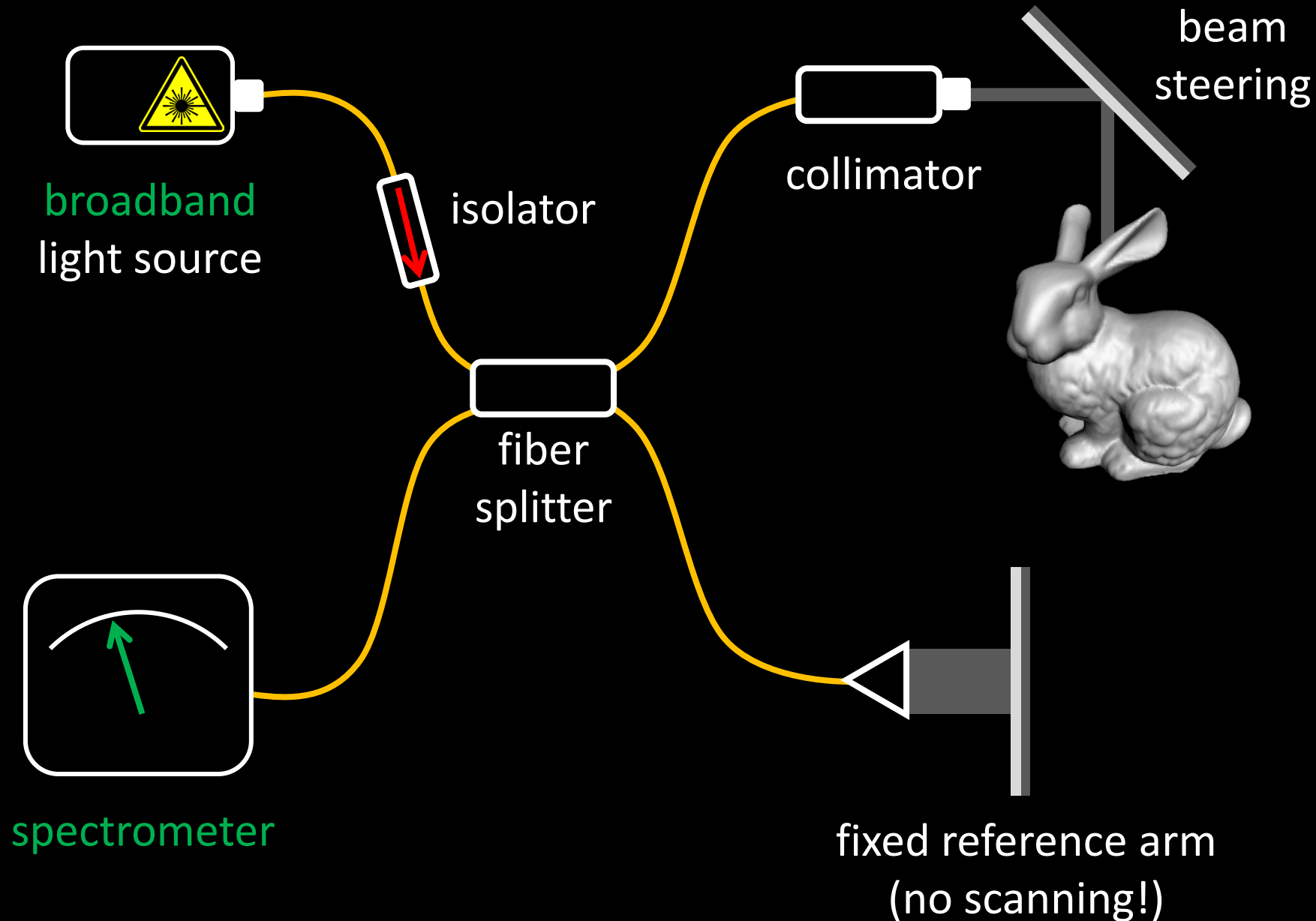
Time-domain optical coherence tomography



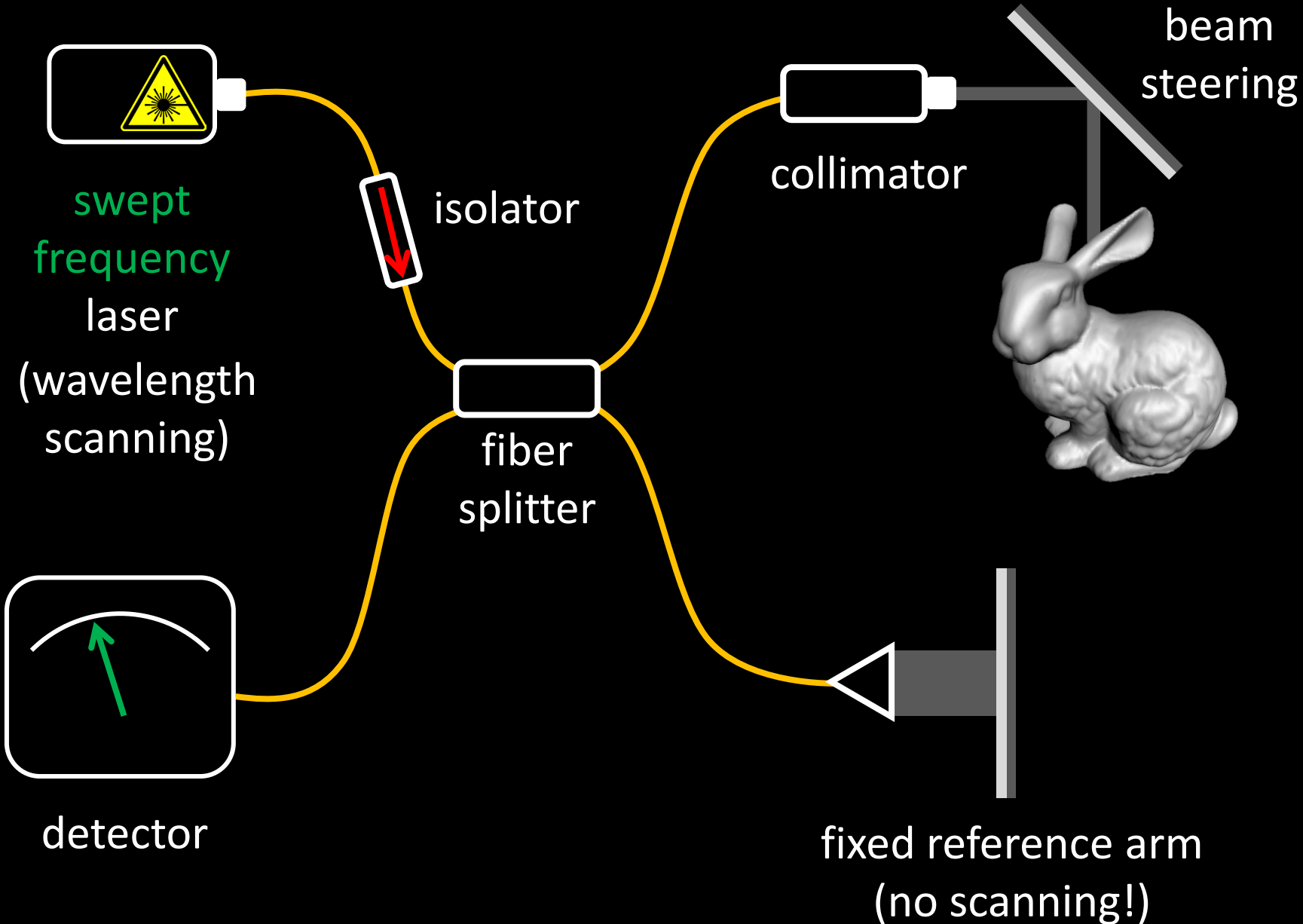
Time-domain optical coherence tomography



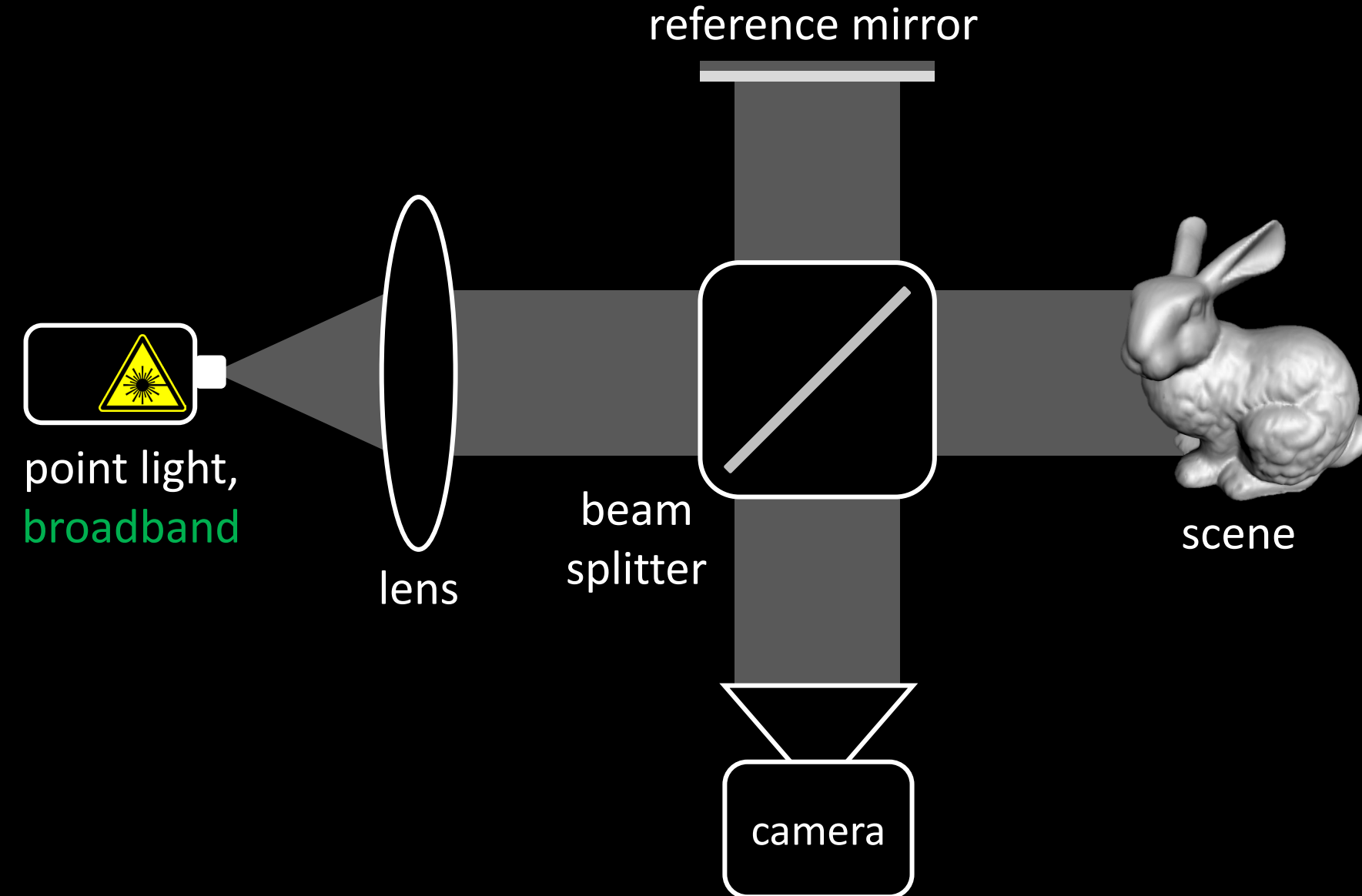
Frequency-domain optical coherence tomography



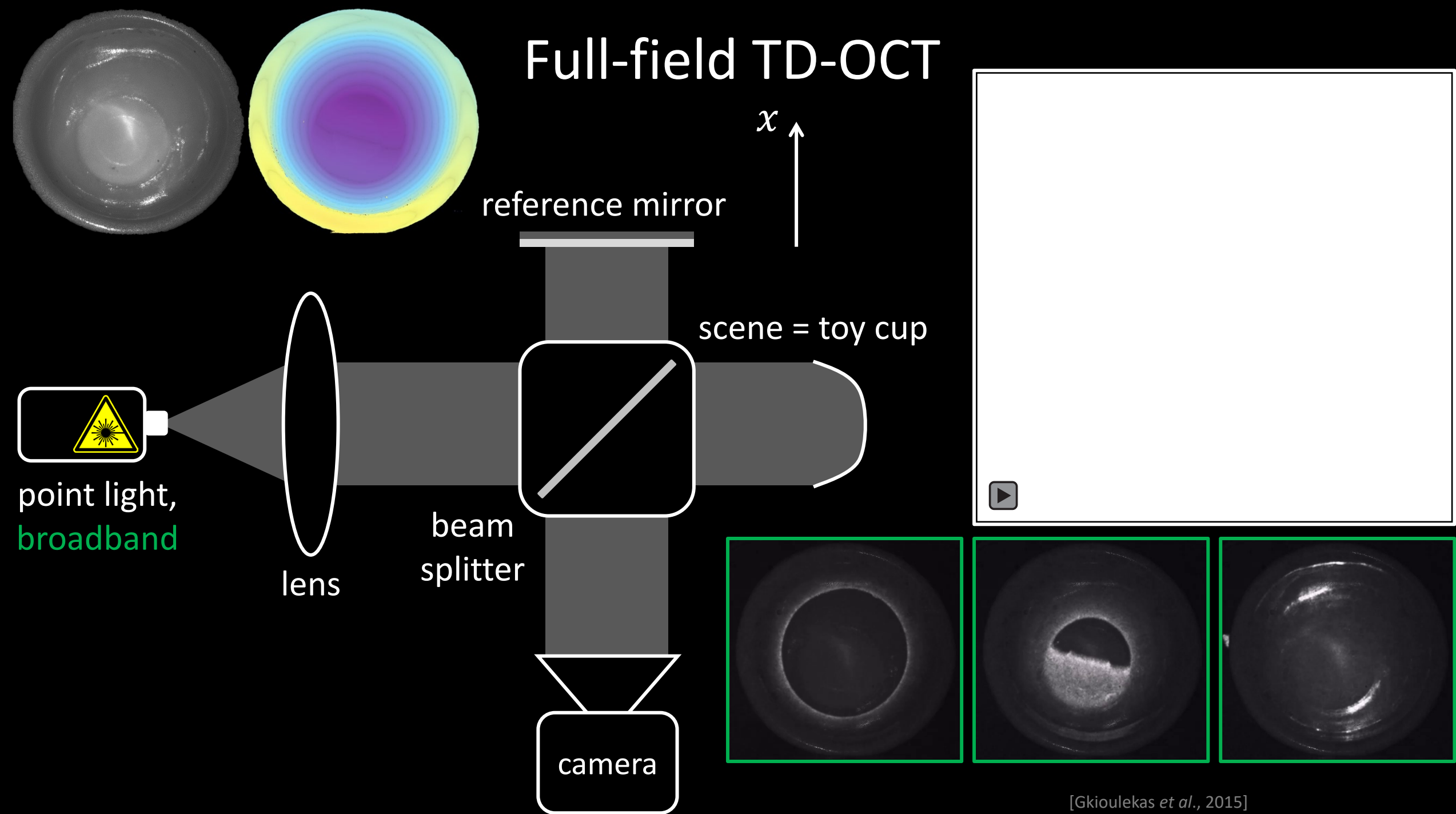
Swept-source optical coherence tomography



Full-field OCT



Full-field TD-OCT



Micron-scale shape scanning

depth resolution $\sim 5\text{ }\mu\text{m}$

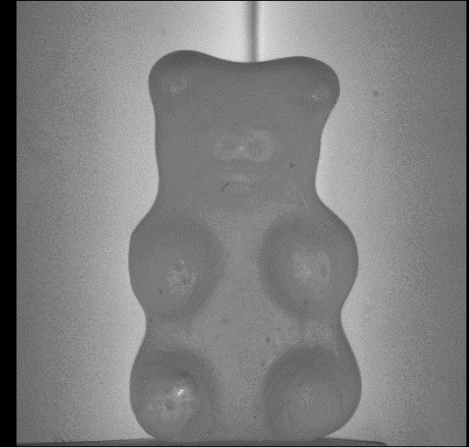
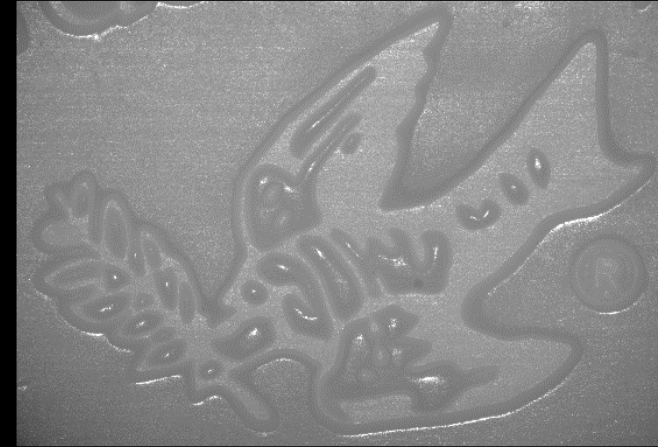
coin

gnocchi

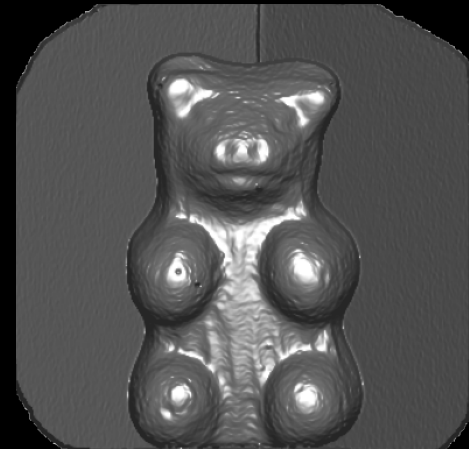
soap carving

gummy bear

captured image



scanned shape



OCT vs SWI

	full-field OCT	full-field SWI

OCT vs SWI

	full-field OCT	full-field SWI
depth range	constrained by reference translation	synthetic wavelength (separation between wavelengths)

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phase wrapping	no	yes

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	full-field OCT	full-field SWI
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depth resolution	spectral bandwidth	separation between wavelengths

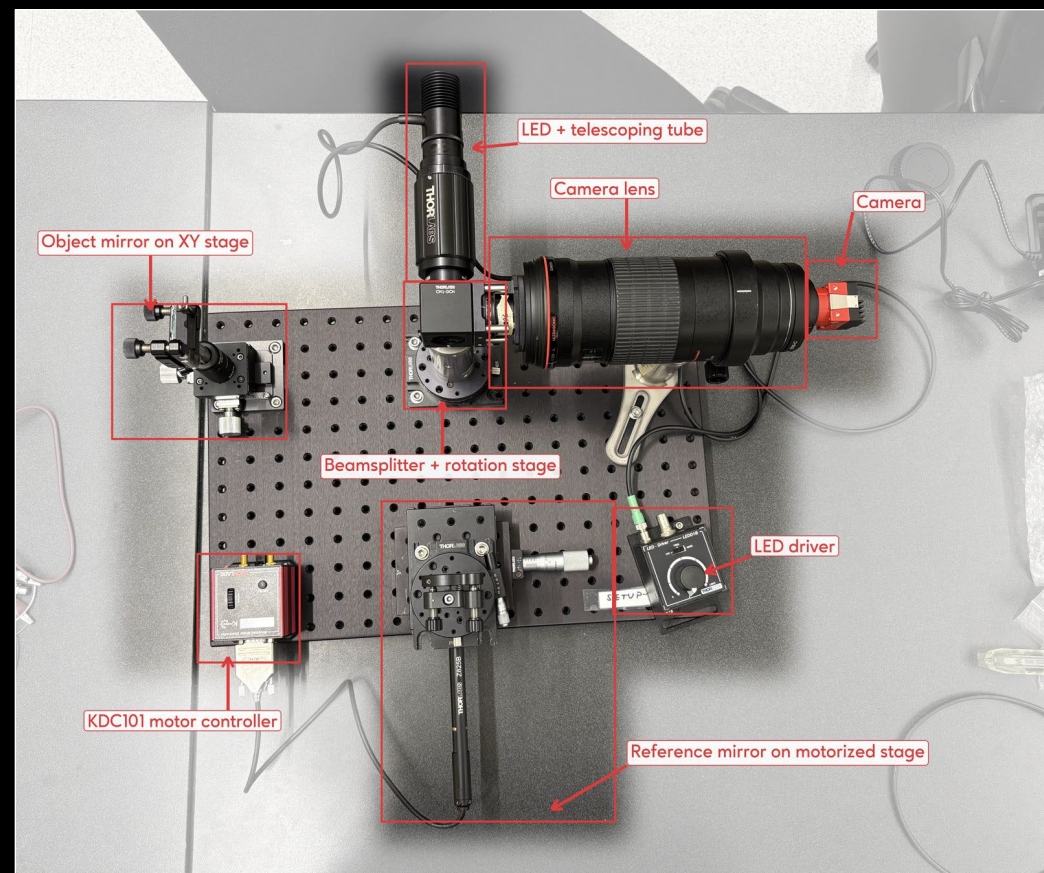
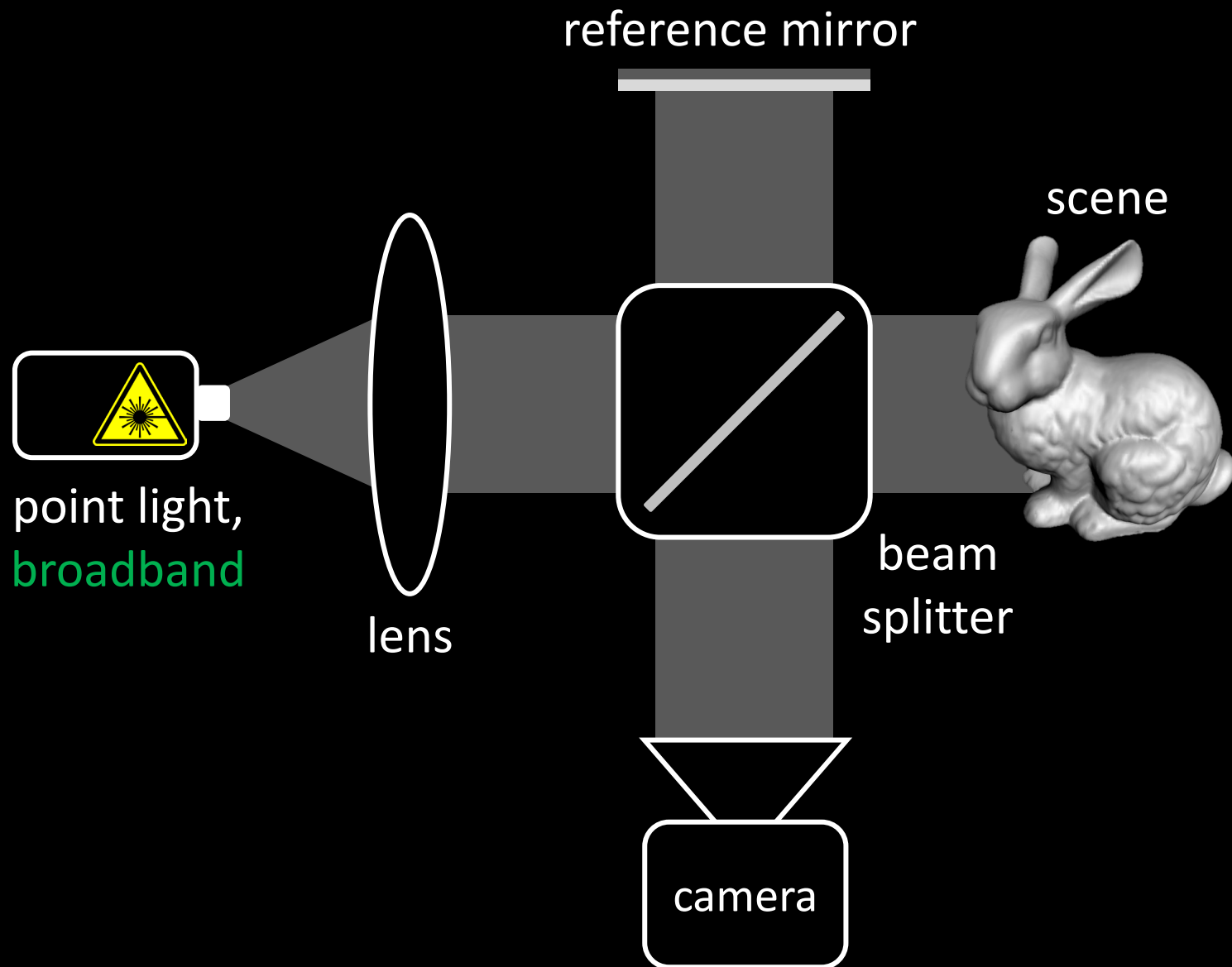
OCT vs SWI

	full-field OCT	full-field SWI
depth range	constrained by reference translation	synthetic wavelength (separation between wavelengths)
phase wrapping	no	yes
depth resolution	spectral bandwidth	separation between wavelengths
scanning	axial proportional to depth range OR lateral proportional to scene size	fixed axial scanning

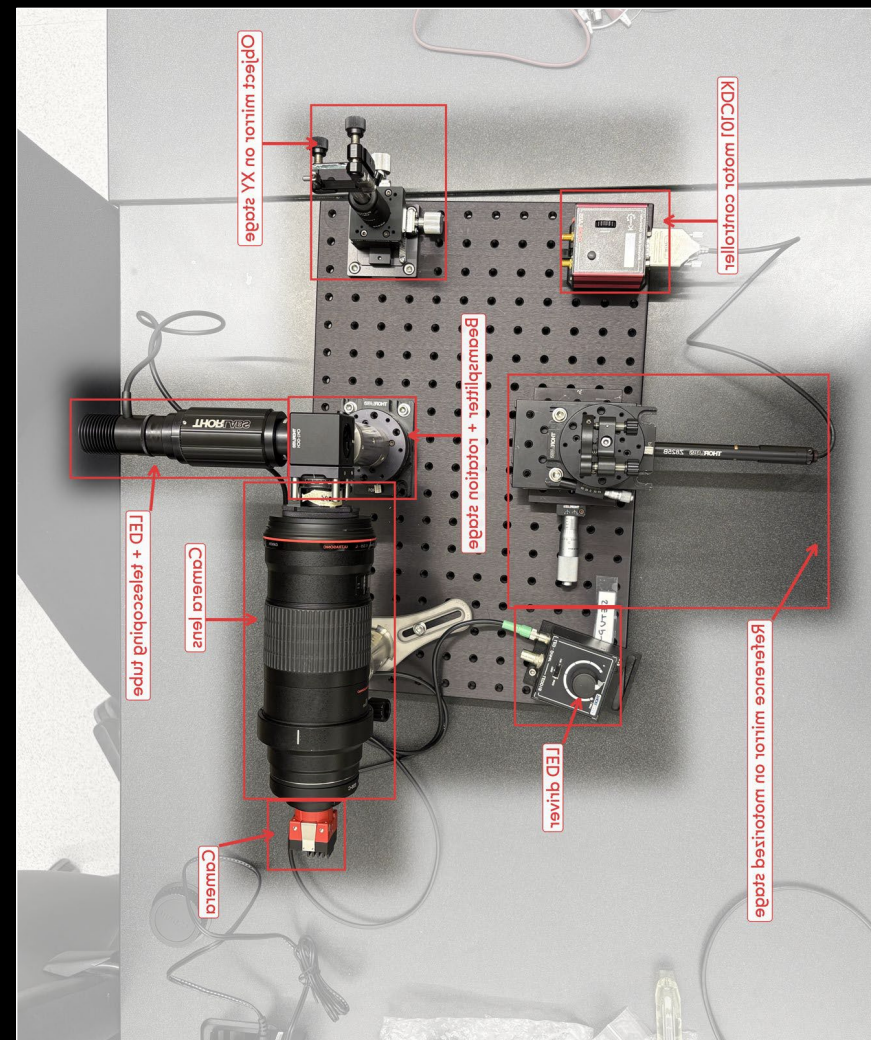
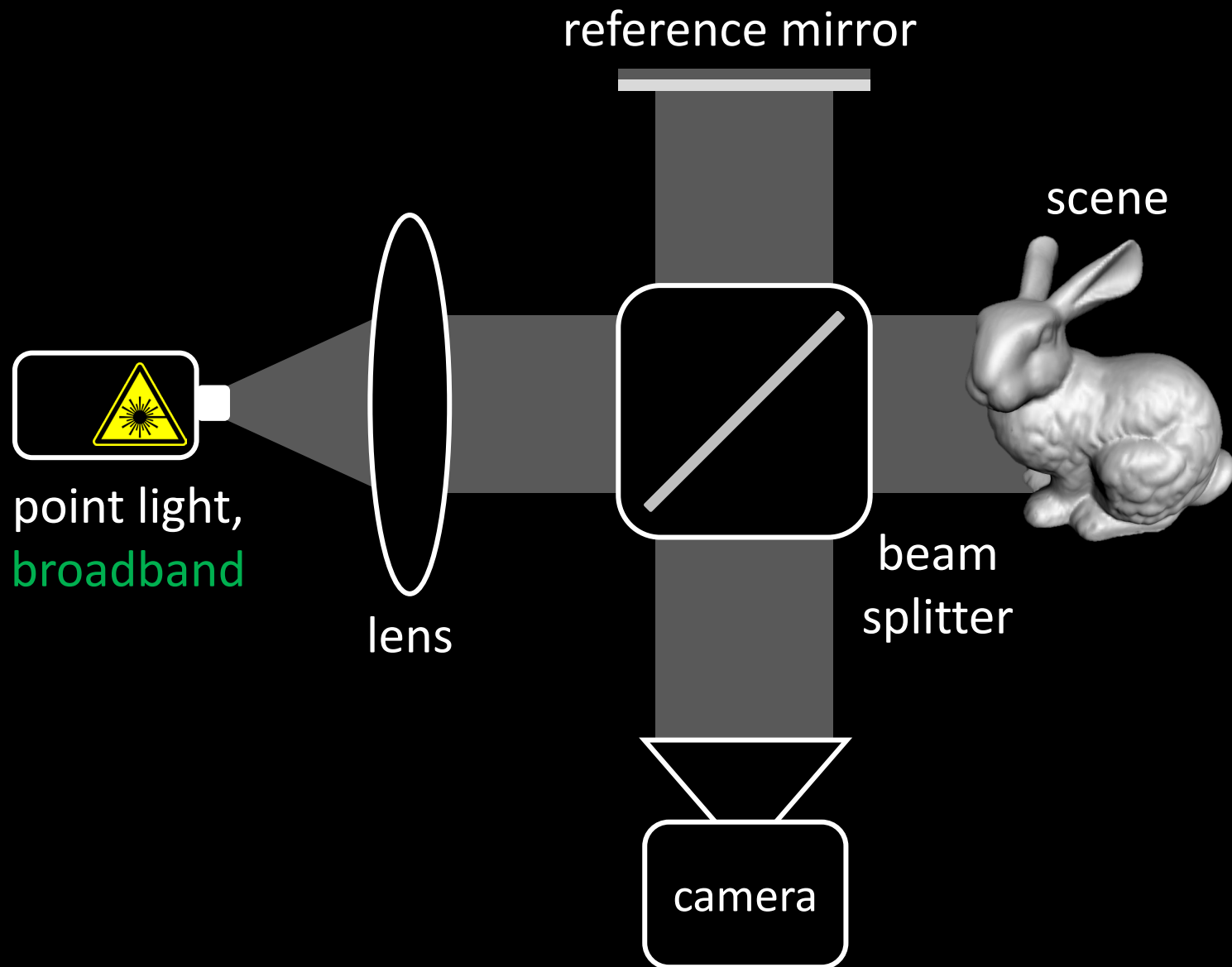
OCT vs SWI

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depth range	constrained by reference translation	synthetic wavelength (separation between wavelengths)
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measurement	full tomography	surface-only

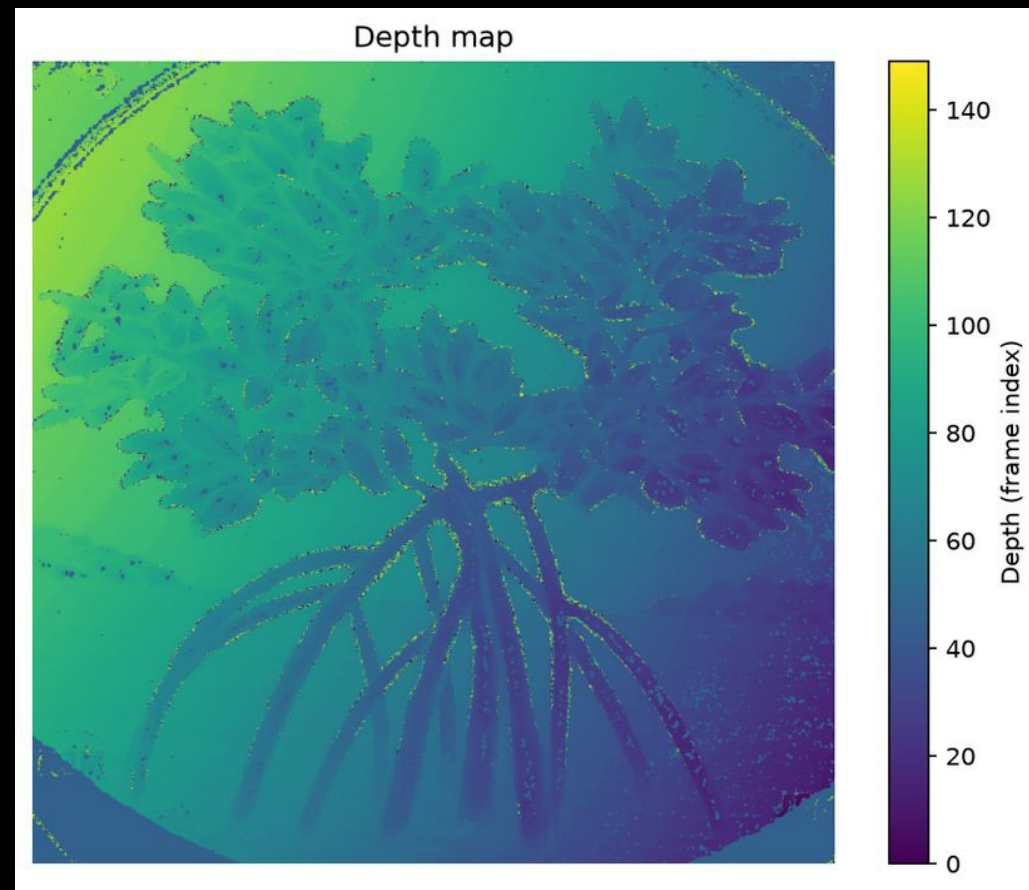
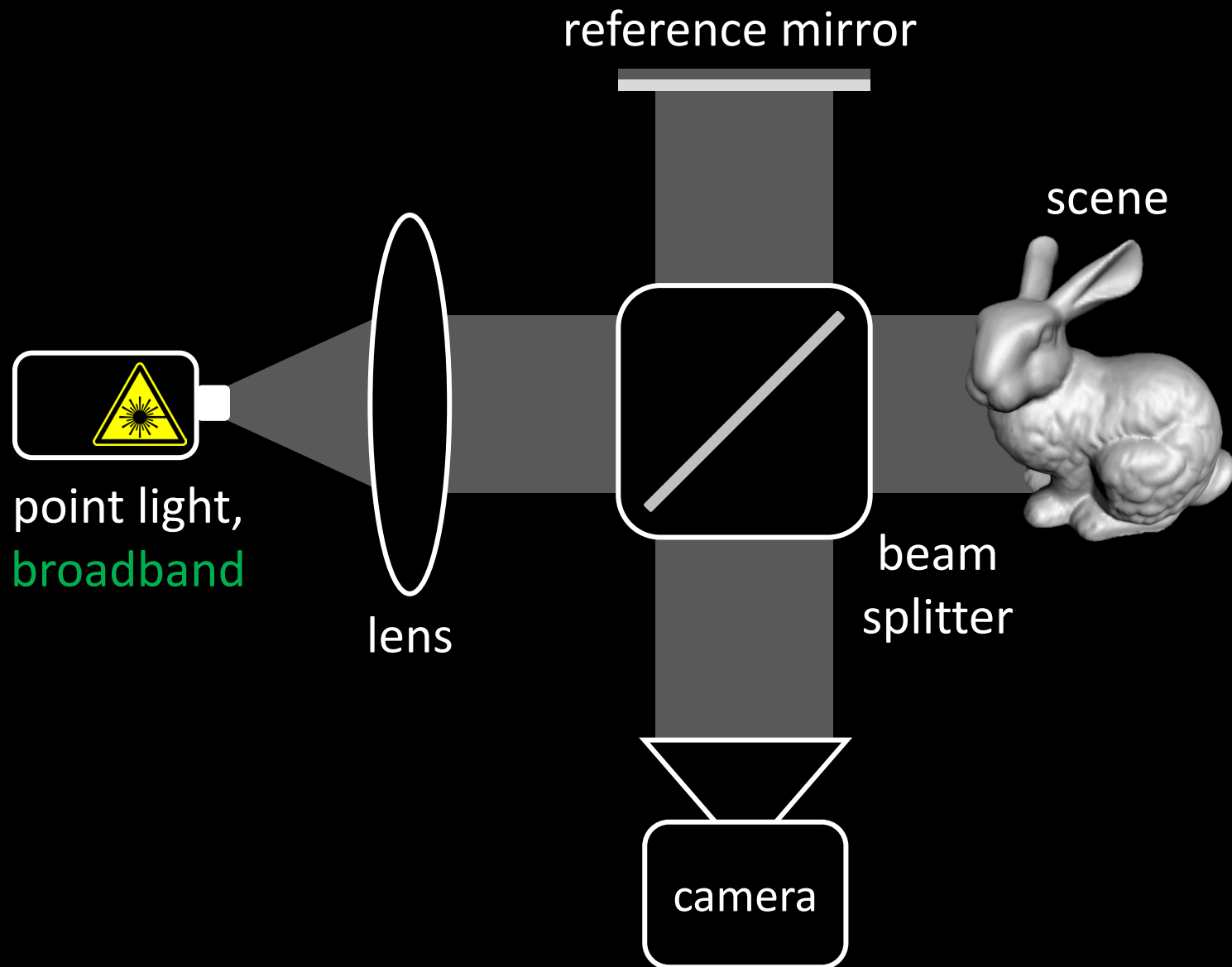
Full-field OCT



Full-field OCT



Full-field OCT



Detailed instructions



<https://tinyurl.com/oct-iccp>

Full-field OCT

